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THE RESPONSE OF BENTHIC COMMUNITIES IN NARRAGANSETT BAY TO CHANGING NITROGEN INPUT AND TEMPERATURE

Shaina Harkins
University of Rhode Island, shainaharkins@uri.edu

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THE RESPONSE OF BENTHIC COMMUNITIES IN NARRAGANSETT BAY
TO CHANGING NITROGEN INPUT AND TEMPERATURE

BY

SHAINA HARKINS

A THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE DEGREE OF

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MASTER OF SCIENCE
OF
SHAINA HARKINS

APPROVED:

Thesis Committee:

Major Professor	Jeremy Collie
	Candace Oviatt
	Jacqueline Webb
	Conor McManus

Brenton DeBeof

DEAN OF THE GRADUATE SCHOOL

UNIVERSITY OF RHODE ISLAND

2023

ABSTRACT

Marine ecosystem changes are often driven by different forces simultaneously, making it challenging to evaluate the individual contributions of each force. In Narragansett Bay, nitrogen input from wastewater treatment facilities (WWTF) was successfully decreased by 50% from 2005 to 2013. In the last 50 years, surface water temperature in the Bay has also increased by 1.6°C. The response of the benthic community to either of these changes has yet to be determined. We replicated the historical methodology of diver collected cores for infaunal macrofauna, organisms living in sediment, and included beam trawl sampling to catch larger, more mobile epifauna, organisms living on sediment. Replicate samples were collected from three stations along the north-south nutrient gradient of the West Passage to understand the effects of nitrogen and temperature on the species composition of the macrofaunal community. Historically, a more advanced successional stage of benthic communities was found further south along the north-south gradient of the Bay's West Passage. We found three distinct communities along the north-south gradient, with the upper Bay community resembling the mid Bay community more over time. Change in nitrogen was identified as the environmental variable with the highest correlation to change in benthic community abundance over time ($\rho = 0.605$), followed by dissolved oxygen ($\rho = 0.605$) and temperature ($\rho = 0.597$). Results suggest high epifaunal biomass and late summer hypoxia from increased rainfall in 2021 caused a decrease in macrofaunal community abundance and biomass, as

well as a responding decrease in epifaunal biomass, in the upper Bay community between years of sampling. The results of the current study reveal the impacts of environmental variables on the benthic community in Narragansett Bay following successful nutrient reduction, which can be used to inform management practices of other temperate estuaries experiencing similar ecosystem changes.

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Science is never accomplished alone, a phrase I continually repeat to remind myself that I am not alone in this uphill battle, I have the support of many.

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Finally, I would like to thank Rhode Island Sea Grant for funding this research.

DEDICATION

I would like to dedicate this to my mom, MJ. When I was little, I was convinced she was a superhero, and that hasn't changed. She successfully raised five children, working endlessly to support us, no matter the circumstances. She somehow always kept her optimism and her head held high for her kids. No matter what mood I call her in, whether it is to stress rant, because I was lonely, or just to say hi, she is always excited to hear from me. She has supported me through every endeavor with the unwavering support of a fantastic mother who has the faith that her daughter can do anything. She was always the main support in my pursuit of science and knowledge, buying me field guides and subscriptions to nature magazines, taking me to zoos and museums, enthusiastically listening to me talk her ear off about cool science facts or my latest animal obsession, and has continued to support me through higher education somehow knowing I would succeed. She said if she hadn't had her kids, she would have been a veterinarian. While I'm grateful she chose the former, I continue to pursue science with her as my main motivation and inspiration, knowing she gave up her passion so I could pursue my own. On my desk at home is a piece of art she gave me that reads, "You are braver than you believe, stronger than you seem, and smarter than you think." When I feel uncertain, I look at that picture, I think of her warmth, and with her in mind, I tell myself, "You can do this." I hope she knows the

infinite love and appreciation I feel for her, and that those words apply to her, too.

PREFACE

This thesis is presented in standard format. The benthic community of Narragansett Bay has been periodically tested amidst a history of decreasing nitrogen input after decreases in wastewater treatment inputs and increasing temperatures from climate change. The intention of this study was to survey benthic macrofauna at three stations throughout the West Passage of the Bay to compare the response of benthic community composition to changing nitrogen input and temperature over time. The macrofaunal data collected in this survey replicated the methods of past benthic surveys and compared it to new data collected. The current study also included the collection of epifaunal data, as well as biomass of both assemblages.

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INTRODUCTION

Narragansett Bay is the largest estuary in southern New England (Figure 1). Tidal forcing pushes Rhode Island Sound water into the lower East Passage, where it mixes with terrestrial, freshwater inputs from the various tributaries of Providence River as it moves northward, eventually being channeled southward and out through the West Passage (Pilson 1985a). Compared to the East Passage, the West Passage is relatively shallow due to glacial deposition from alluvial fans and fan deltas (Boothroyd & August 2008). Mixing from tidal forcing and relatively shallow depths in estuaries contribute to strong coupling between benthic and pelagic ecological processes and nutrient cycling (Doering 1989; Pilson 1985b).

Anthropogenic development of the Bay began in the 17th century, steadily increasing through the 20th century. The 20th-century accelerated industrial development, disrupting a highly diverse and productive ecosystem with intense fishing pressure, pollution and sewage runoff (Oviatt et al. 2003; Nixon 1991; Nixon et al. 2008). Sewage nutrient input increased to a maximum in the mid-20th century, then decreased due to improved wastewater treatment (Nixon et al. 2008). The most densely populated area surrounding the Bay is the city of Providence, bordering the upper Bay. It is also the area with the highest contribution of pollution, sewage, impervious services, and freshwater runoff to the Bay. Decreasing gradients of human impact, nutrients, and freshwater input exist from the upper Bay to the less-densely populated lower Bay (Oviatt 2008).

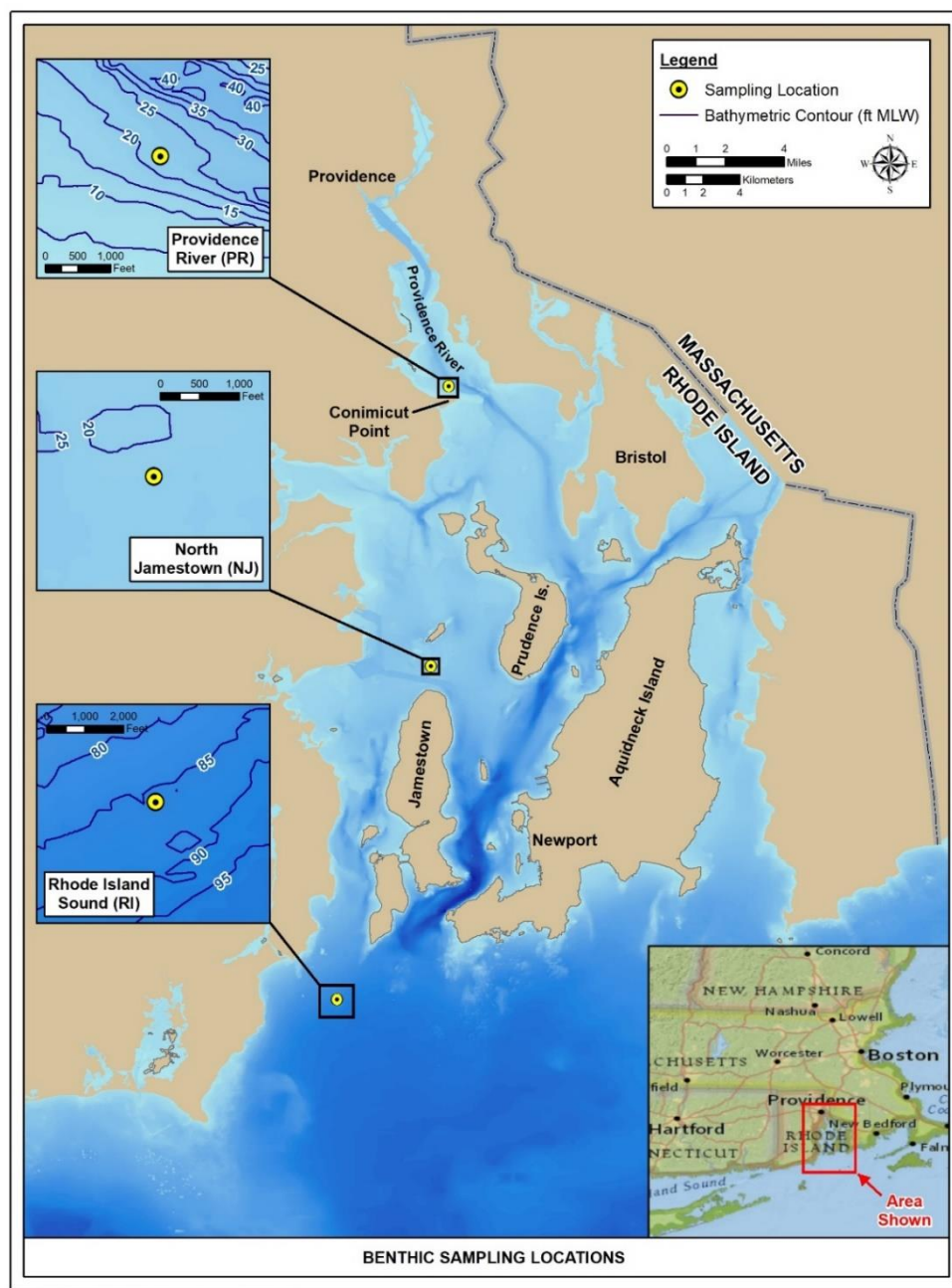


Figure 1: Map of Narragansett Bay station locations with benthic sampling locations. Magnified subplots of stations with depth contours are also presented.

NITROGEN AND TEMPERATURE

Regular Bay-wide monitoring of nitrogen input from sewage and river discharge began in the 1970s (Nixon et al. 2008). Total nitrogen (TN) load from river discharge and wastewater treatment facilities (WWTF) in 1982-1983 was estimated to be 8131×10^3 kg/year (Nixon et al. 2008, Oczkowski et al. 2018). In 2013-2015, TN load from river discharge and WWTF decreased to an estimated 3570×10^3 kg/year (NBEP 2017, Oczkowski et al. 2018).

In the early 2000s, efforts by the US Environmental Protection Agency and the Rhode Island Department of Environmental Management were underway to reduce nitrogen from WWTF into the Bay. In August 2003, Greenwich Bay, a sub-estuary of upper Narragansett Bay, experienced a localized anoxic event triggering a fish and invertebrate die-off (RIDEM 2003). Eutrophication induced by nitrogen loading from WWTF lacking tertiary nitrogen removal, in combination with temperature stratification and weak neap tide conditions, caused the upper Bay anoxia event (Deacutis et al. 2006). Following the anoxic event, total nitrogen input was reduced by 50-60% between 2005 and 2013 for all WWTF discharging to Narragansett Bay to combat prolonged eutrophication and hypoxia (Oviatt et al. 2017, Oczkowski et al. 2018). The nitrogen reduction in the upper Bay was anticipated to decrease the degree of excess accumulation of organic matter that fuels increased benthic respiration, leading to an increase in dissolved oxygen near the benthos (Nixon et al. 2008), with the possible ecosystem effect of decreased productivity in the lower Bay (Nixon et al. 2008).

As nitrogen input reductions to the Bay were underway in an effort to decrease eutrophication, climate was concurrently changing. Average sea surface temperature in the Bay increased 1.6°C in the last 50 years (Fulweiler et al. 2015). Both nitrogen-induced eutrophication and temperature stratification contributed to the Greenwich Bay anoxia in 2003 (Deacutis et al. 2006). Increased sea surface temperature and rainfall were anticipated to promote further density-driven stratification in the upper Bay, which would increase the risk of bottom water hypoxia. Increased bottom water temperature is expected to increase metabolism of benthic organisms, and therefore rates of oxygen consumption, further increasing the risk of hypoxia (Fulweiler & Nixon 2009).

BENTHIC ORGANISMS

At the beginning of tertiary nitrogen removal from WWTF in the early 2000s, a study on the response of macrofauna to anthropogenic stressors in Narragansett Bay was performed (Calabretta & Oviatt 2008). Diver-collected core samples between June of 2000 and 2004 for three stations in the upper Bay, Conimicut Point, Greenwich Bay, and Mount Hope Bay, were compared to a control site north of the island of Jamestown, collected between June 2000 and June 2010. The comparison assessed the impact of heavy metal contaminants and nutrient loading to benthic communities along the Bay's north-south pollution gradient. North of Jamestown benthic community assemblage changed little between years, but the benthic community

assemblage of the three upper Bay stations were similar between stations and within years. The upper Bay community changed as a whole year-to-year, with the colonization of opportunistic species each year, while the benthic community at the site north of Jamestown remained relatively stable (Calabretta & Oviatt 2008).

Benthic macrofauna have long been considered useful bio-indicators of ecological quality in shallow marine ecosystems. They typically react to disturbance in a predictable manner, demonstrating a range of stress tolerances and trophic interactions (Weisberg et al. 1997). A successional response of benthic macrofauna exists at pollution sources such as WWTF (Pearson & Rosenberg 1978; Rhoads et al. 1978). At the pollution source, most macrofaunal organisms will be small in size and number. In Narragansett Bay, the most ubiquitous and tolerant benthic macrofaunal species is a small capitellid polychaete, *Mediomastus ambiseta* (Calabretta & Oviatt 2008). Downstream from the pollution source, organism abundance increases to a peak, revealing an assemblage dominated by opportunistic species. Opportunistic assemblages are often dominated by high abundance, low biomass, tolerant tube-dwelling polychaetes and amphipods with high fecundity, feeding at the sediment surface (Pearson & Rosenberg 1978; Rhoads et al. 1978). *Mediomastus ambiseta* is also found in low successional areas, joined by dense mats of ampeliscid amphipods and spionid polychaetes, *Streblospio benedicti* and *Polydora cornuta* (Calabretta & Oviatt 2008). Further from a disturbance species richness, calculated from the number of species

present in an assemblage, species evenness, the relative abundance of individuals among each species, and biomass increase as abundance decreases. Overall abundance will decrease to a steady-state, with the community dominated by high biomass, deeper-dwelling organisms with a longer lifespan and lower fecundity (Pearson & Rosenberg 1978; Rhoads et al. 1978). *Nephtys incisa* and *Nucula proxima* are the dominating species of high successional stage assemblages of the Bay (Calabretta & Oviatt 2008).

Narragansett Bay's benthic macrofaunal community is an important food source to fishes and a source of recycled nutrients. Groundfish species such as winter flounder (*Pseudopleuronectes americanus*) feed on small, opportunistic tube-dwelling polychaetes (Meng et al. 2001). Opportunistic, surface-dwelling macrofauna are highly productive with high fecundity. They grow rapidly, with the ability to regenerate after non-lethal predation, suggesting a renewable, fast growing, readily available food source (Grassle & Grassle 1984). Recycled nutrients from the benthos account for a significant portion of nutrients required for annual primary production (Nixon 1981).

SAMPLING METHODOLOGY USED IN PRIOR STUDIES

Many benthic sampling protocols are limited in time, space, and methodological replication. Periodic, long-term benthic sampling is laborious and expensive. Sampling technique and station location have varied among studies of the macrofauna of Narragansett Bay. With respect to the sampling of macrofauna alone, methodology and sieve size have lacked consistency

(Table 1), which has prevented quantitative comparison of results. Diver-collected cores preserve the flocculent sediment-water interface, while grab samples and clam shell buckets are not as delicate (Calabretta & Oviatt 2008). Diver-collected cores, while providing more detailed sampling of macrofauna, almost completely exclude large epifauna. To fully encapsulate the benthic community, a sound macrofaunal sampling technique should be coupled with independent epifaunal sampling at the same station. Only a few studies used replicated techniques and stations, bolded in Table 1. However, studies using different methodologies can be used for qualitative comparisons of major species presence/absence and metrics of biodiversity at stations through time (Hale et al. 2018a).

Table 1: Historical datasets for comparison. A compilation of historical datasets to compare to current data. Bolded datasets employed diver collected cores. RI = Rhode Island Sound station, NJ = site north of Jamestown station, PR = Providence River station. *Data for NJ 2000-2010, data for PR 2000-2004.

Dates/ Interval sampled	Stations sampled	Sampling method	Area sampled (cm ²)	Smallest sieve size (µm)	Study Number [†]
3/1957	RI	Clam shell bucket	2000	500	1
8/1967	PR, NJ, RI	Van Veen grab	1000	500	2
11/1974	NJ	Smith-McIntyre grab	1000	750	3
7/1975	PR	Smith-McIntyre grab	400	750	4
7/1975-7/1976 quarterly	PR, NJ	Diver collected cores	175/322	500	5
8/1977-8/1978 quarterly	NJ	Diver collected cores	420	300/500	6
6/1977-5/1979 quarterly	NJ	Diver collected cores	35.3	300/500	7
8/1983-4/1986 monthly	NJ	Diver collected cores	17.35	300/500	8
6/1990- 9/1990 monthly	PR, NJ, RI	Smith-McIntyre grab	470	500	9
6/2000-6/2010 yearly*	PR, NJ	Diver collected cores	8.04	300/500	10
6/2021-6/2022 yearly	PR, NJ, RI	Diver collected cores	19.6	500	11

[†]Study: 1. Phelps 1958, 2. Chowder & Marching Jiffy Cruise 1967, 3. Hale 1974, 4. Pratt & Bisagni 1976, 5. Myers & Phelps 1978, 6. Hyland 1981, 7. Rudnick 1984, 8. Hughes 1986, 9. French et al. 1992, 10. Calabretta & Oviatt 2008, 11. Current study (see Chapter 2: Methodology and Procedures)

OBJECTIVES

The current study will analyze the differences in macrofaunal and epifaunal community composition between north, mid, and south stations in the West Passage of Narragansett Bay to assess the disturbance level throughout the Bay. These data will then be compared to historical data obtained using the same sampling methods in order to analyze the impact of environmental variables. It is hypothesized that changes in environmental variables through time have changed the macrofaunal community composition of the upper Bay. The objectives of the current study are 1. to assess the current macrofaunal community assemblage of the Bay throughout the environmental gradient of the Western Passage at north, mid, and south stations, 2. to assess the epifaunal benthic community assemblage of the Bay at the same three stations 3. to assess the effect of the change in nitrogen, temperature, and related environmental variables through time on the change in macrofaunal community assemblage from the peak of nitrogen input in the late 1970s and mid-1980s, through the beginning of the decrease in WWTF nitrogen input in the early 2000s, to current, 2021-2022, 7-8 years after the decrease in WWTF nitrogen input was completed.

METHODOLOGY AND PROCEDURES

SAMPLING SITES

We collected samples for the current study from three stations throughout Narragansett Bay on June 29th and 30th of 2021 and June 15th and 29th of 2022. Sampling stations were selected to mirror previously sampled stations and employ the same methodology of diver collected cores along the Bay's north-south nutrient gradient (Table 1, Figure 1). The Providence River (PR) station (41° 43.4', 71° 21.7') was the northern-most station, with the shallowest depth, highest bottom temperature, and lowest salinity of the three stations (Table 2). The Providence River station was directly comparable to cores collected in 1978 and 2008 (Myers & Phelps 1978, Calabretta & Oviatt 2008, Table 1). The North of Jamestown (NJ) station (41° 35.0', 71° 22.2') represented the mid-Bay station, with a greater depth, lower bottom temperature, and higher salinity than the Providence River station (Table 2). The North of Jamestown station has been the most frequently sampled site in Narragansett Bay, with data directly comparable to those collected in: 1975-1979, 1983-1986, and 2000-2010 (Table 1). The Rhode Island Sound (RI) station (41° 25.0', 71° 25.0') was the southern-most station located outside the Bay. This station was the deepest channel, lowest bottom temperature, and highest salinity of the three stations sampled (Table 2). Because the collection methods were not the same as previous studies (Table 1), the current Rhode Island Sound dataset was not directly comparable to historical stations sampled, preventing temporal comparisons of the station.

Table 2: Current environmental data. Collected during the current study from YSI measurements at the time of sampling and sediment analysis from Boston University Coastal Ecology and Biogeochemistry Lab.

Metric	PR 2021	NJ 2021	RI 2021
Depth (m)	8.5	20.5	26.25
Bottom Temperature (°C)	19.8	18	12.61
Salinity	30.6	32	33.53
Dissolved Oxygen (%)	62.1	85.8	71.1
Dissolved Oxygen (mg/L)	4.71	6.61	6.21
Sediment Nitrogen (%)	0.36	0.17	0.075
Sediment Carbon (%)	3.67	1.64	0.8
Sediment Organic Carbon (%)	9.32	5.15	2.57
Metric	PR 2022	NJ 2022	RI 2022
Depth (m)	7.66	20.65	26.77
Bottom Temperature (°C)	18.3	17.16	12.14
Salinity	30.15	31.19	31.98
Dissolved Oxygen (%)	70.7	91.9	86.3
Dissolved Oxygen (mg/L)	5.85	7.17	7.3
Sediment Nitrogen (%)	0.36	0.17	0.09
Sediment Carbon (%)	3.96	2.05	0.87
Sediment Organic Carbon (%)	9.8	5.86	2.15

MACROFAUNAL SAMPLE COLLECTION AND ANALYSIS

Each year, 15 replicate sediment cores were hand-collected at each station by the URI-GSO dive team. Cores were obtained in June of 2021 and 2022 for comparison to previous studies that identified June as the peak abundance season for most benthic invertebrates in Narragansett Bay (Rudnick 1984, Calabretta & Oviatt 2008). Cores consisted of 5-cm diameter and 30-cm long acrylic tubes that were carefully inserted into the sediment to a depth in excess of 10 cm, and closed at each end *in situ* with rubber stoppers to ensure the preservation of the flocculent sediment-water interface. Cores were processed immediately upon arrival on deck, the overlying water and top 2 cm of sediment were stored in one container and the bottom 8 cm were stored in a separate container as per methods in previous studies (Elmgren 1973, Ellis 2002, Calabretta & Oviatt 2008). The layers of sediment were separated to streamline the sorting process. Between 50 and 85% of organisms were known to reside in the top 2 cm, as compared to the more hypoxic bottom 8 cm of sediment (Sanders 1960). Samples were fixed with 4% pH buffered formalin and stained with rose Bengal to ease sorting and identification of invertebrates. In the lab, samples were carefully rinsed through a 500- μ m sieve, the minimum size defining adult macrofauna, as in prior studies in Narragansett Bay (Frithsen 1989, Ellis 2002, Calabretta & Oviatt 2008). Organisms were identified to the lowest possible taxonomic level using various resources (Abbott 1954, Pittibone 1963, Smith 1964, Gosner 1971, Bousfield 1973, Brinkhurst et al. 1975, Fauchald 1977, Gosner 1978, Appy

1980, Weiss 1995, WoRMS 2023), counted, and, once enumerated, specimens were stored in 70% ethanol. Abundance of each species per replicate was divided by core area (0.00196 m^2) to convert counts to number of individuals per m^2 .

Another core was collected at each station to extract 10-mL subsamples of sediment with a modified syringe. These were shipped to the Coastal Ecology & Biogeochemistry Lab at Boston University (BU) for analysis to measure carbon and nitrogen content at 1-cm depth increments. The top 10 cm of sediment along the West Passage of Narragansett Bay are considered an 'enriched zone' due to mixing from bioturbation, leading to little variation in the values of both percent carbon (%C) and percent nitrogen (%N) throughout the depth increments (Nixon 1986). The %C and %N values were, therefore, averaged over the 1-cm increments within the top 10 cm for each station in each year. Physical data were recorded at each station with a calibrated YSI Sonde 6920, including surface and bottom temperature, salinity, and dissolved oxygen measurements.

EPIFAUNAL SAMPLE COLLECTION AND ANALYSIS

Two replicate, 15-minute beam trawl tows were performed at each core station to estimate epifaunal abundance. The beam trawl consisted of a 1.5-cm mesh size net, with a 0.5-cm in-seam liner, attached to a 2-m wide metal beam and two metal skis that kept the net open as it dragged along the bottom collecting epifauna. Larger organisms were handled on board, where they

were identified, counted, weighed with an electronic balance to the nearest gram, and length measured with a measuring board to the nearest millimeter before being released. Smaller organisms were preserved and lab-processed in the same manner as the macrofaunal sediment core samples; sieved through a 500- μm sieve, sorted, identified to the lowest possible taxonomic level, counted, and stored in 70% ethanol. Average boat speed of each replicate tow and tow time were recorded to calculate replicate tow distance. A sampling area was calculated for each replicate by multiplying the tow distance by 2 m, the beam width. To convert abundances to number of individuals per m^2 , the abundance per replicate for each species was divided by associated sampling area.

HISTORICAL MACROFAUNAL ABUNDANCE DATA

Historical studies employing the same methods and stations were compiled (Myers & Phelps 1978, Hyland 1981, Hughes 1983, Rudnick 1984, Calabretta & Oviatt 2008, Table 1). For studies that considered multiple size fractions, only macrofauna greater than 500 μm were included (Table 1). Abundance data were averaged across replicates for each year sampled, and converted into abundance per m^2 . Analysis included only data from months of May-September, representing summer peak abundance for the benthic community (Rudnick 1984). Due to the low number of species caught, two outlier samples, a PR 1975 core and a NJ 1977 core, were omitted (Myers & Phelps 1978, Hyland 1981).

COMPARISONS

Two spatial and three temporal comparisons were performed to fulfill the objectives of the current study (Table 3):

1. **Current macrofaunal comparison:** a comparison among the three stations and two years of data collected in the current study to assess the spatial differences in current macrofaunal community composition against nitrogen and temperature.
2. **Current epifaunal comparison:** a comparison among the three stations and two years of data collected in the current study to assess the spatial differences in current epifaunal community composition against nitrogen and temperature.
3. **Before/after tertiary nitrogen macrofaunal comparison:** a temporal comparison among the NJ and PR stations, before and at the beginning of tertiary nitrogen removal implementation to WWTF, from 2000-2004, and after completion, from 2021-2022, against nitrogen and temperature.
4. **Historical core macrofaunal comparison:** a temporal comparison among the current NJ and PR station macrofaunal abundance and the directly comparable historical diver-collected core macrofaunal abundance data against nitrogen and temperature (Table 1, bolded).
5. **Historical full macrofaunal comparison:** the change in taxonomic distinctness among macrofaunal abundance studies through time at each station, regardless of sampling methodology (Table 1, all).

Table 3: Comparisons made in the current study. NBFSMN – Narragansett Bay Fixed Site Monitoring Network, NBFTS – Narragansett Bay Fish Trawl Survey, NBLTPTS – Narragansett Bay Long Term Plankton Time Series.

Comparison	1	2	3	4	5
Current Benthic Stations	PR, NJ, RI	PR, NJ, RI	PR, NJ	PR, NJ	PR, NJ, RI
Years	2021-2022	2021-2022	2000-2002, 2004-2008, 2010, 2021-2022	1976-1978, 1983-1985, 2000-2002, 2004-2008, 2010, 2021-2022	1957, 1967, 1975-1979, 1983-1985, 1990, 2000-2002, 2004-2008, 2010, 2021-2022
Analyses	ANOSIM, MDS, SIMPER, BEST	ANOSIM, MDS, SIMPER, BEST	ANOSIM, MDS, SIMPER, BEST	ANOSIM, MDS, SIMPER, BEST	Taxonomic distinctness
Environmental Covariates	sediment carbon, sediment nitrogen, summer (May-September) average temperature, salinity, DO, epifaunal biomass	sediment carbon, sediment nitrogen, summer (May-September) average temperature, salinity, DO	yearly average (September previous year-August) total DIN, NH ₄ , NO ₃ +NO ₂ , summer (May-September) average temperature, salinity, DO, chlorophyll a	yearly average (September previous year-August) total DIN, NH ₄ , NO ₃ +NO ₂ , summer (May-September) average temperature, salinity, chlorophyll a	
Environmental Sources	BU analysis of subsampled cores, NBFSMN, NBFTS	BU analysis of subsampled cores, NBFSMN, NBFTS	NBLTPTS, NBFSMN, NBFTS	NBLTPTS, NBFTS	
Environmental Sites	Current Benthic Stations, Bullock Reach, Conimicut Point, Quonset Point, Fox Island, Whale Rock	Current Benthic Stations, Bullock Reach, Conimicut Point, Quonset Point, Fox Island, Whale Rock	Bullock Reach, Conimicut Point, Quonset Point, Fox Island	Fox Island	

ENVIRONMENTAL DATA

Environmental data were pooled, based on availability, from the Narragansett Bay Long-Term Plankton Time Series (NBLTPTS) (Smayda et al. 1959-1997, NBLTPTS 1997-2022), the Narragansett Bay Fixed Site Monitoring Network (NBFSMN) (Stoffel 2001-2003, NBFSMN 2004-2022), and the Narragansett Bay Fish Trawl Survey (NBFTS) (URI GSO 1959-2022) as their data mirror the timeline (1975-2022) and stations (upper Bay: Bullock Reach and Conimicut Point = Providence River, mid Bay: Fox Island and Quonset Point = North of Jamestown, lower Bay: Whale Rock = Rhode Island Sound) of the current study (Table 3).

All bottom nitrogen data were averaged from the Fox Island station of the NBLTPTS (1975-2022), averaged yearly between September of the previous year and August of the current year to encompass the accumulation of organic waste from the winter-spring bloom (Pilson 1985b). Nitrate (NO_3), nitrite (NO_2), and ammonium concentration (NH_4) data were pooled as total dissolved inorganic nitrogen (DIN) for comparison, as well as compared separately as $\text{NO}_3 + \text{NO}_2$, and as NH_4 .

Bottom temperature and salinity data for the temporal macrofaunal core comparison were pooled from the Fox Island station of the NBLTPTS (Table 3). Bottom temperature, salinity, and dissolved oxygen concentration data for the PR station of the before/after tertiary nitrogen removal comparison were pooled from the Bullock Reach (2000-2003) and Conimicut Point (2004-2022) stations of the NBFSMN. Data for the NJ station of the before/after tertiary

nitrogen removal comparison were pooled from the Quonset Point (2004-2022) station of the NBFSMN (Table 3). Current macrofaunal and epifaunal comparison PR bottom temperature, salinity, and dissolved oxygen concentration data were pooled from the Bullock Reach (2021-2022) station of the NBFSMN. Data for the current NJ macrofaunal and epifaunal comparison were pooled from the Quonset Point (2021-2022) station of the NBFSMN. Data for the current RI macrofaunal and epifaunal comparison were pooled from the Whale Rock (2021-2022) station of the NBFTS (Table 3). Instantaneous YSI data recorded at the time of sampling were also pooled for the current macrofaunal and epifaunal comparisons. All temperature, salinity, and dissolved oxygen data were summer averaged between May and September.

STATISTICAL ANALYSES

The top 2 cm and bottom 8 cm sediment depth horizons of macrofaunal core abundance were combined for analysis of both current and historical data. For each replicate at all stations and years of the current macrofaunal and epifaunal data, total abundance, number of species, biomass, and Pielou's species evenness were calculated. RStudio was used to plot box plots of each metric at each station in each year. Top five species by abundance and biomass were also identified for all stations and years of epifaunal and macrofaunal data.

Plymouth Routines in Multivariate Ecological Research (PRIMER) 6 was used for statistical analysis. For comparisons 1, 2, 3, and 4, abundance data for each station and year were fourth-root transformed to reduce the influence of dominant taxa and Bray-Curtis similarities (Bray & Curtis 1957) were calculated prior to analysis. For comparisons 3 and 4, abundance data were aggregated to the taxonomic family level to decrease identification bias.

Analysis of similarities, or ANOSIM, is a non-parametric significance test between stations and years comparing replicate differences within samples to differences between samples to show if differences between samples were significant (Clarke 1993). Analysis of similarity (ANOSIM) was used to statistically analyze the significance of dissimilarity between stations and years. Comparisons 1, 2, 3, and 4 employed ANOSIM (Table 3).

Non-metric multidimensional scaling (MDS) ordinations allowed for visualization of relationships in benthic community composition between, and within, stations and years (Mead 1992). MDS plots were overlaid with the results of cluster dendrograms, visually displaying similarities between samples at chosen percentages. Multi-dimensional scaling (MDS) was used to plot the comparison of current macrofaunal community similarities and differences between stations, years, and replicates within core samples. Replicates were included to show how within-station abundance varied and clustered. It is important to note that the axes of MDS plots do not hold numeric value, rather, the plot is rendered as a way to visualize the relative similarities and differences in species composition between each replicate

sample in two dimensions. Colored ellipses surrounding clusters indicate percent similarity within those clusters. Comparisons 1, 2, 3, and 4 employed MDS plots (Table 3).

The five species contributing the most to dissimilarity between pairwise stations were obtained from similarity percentages breakdown (SIMPER), as were the five species contributing the most to dissimilarity between years at the Providence River station. Similarity percentages breakdown, or SIMPER, assessed the average percent contribution of species to the dissimilarity to show which species contributed the most to changes between stations and years (Clarke 1993). Comparisons 1, 2, 3, and 4 employed SIMPER (Table 3).

BIO-ENV + BVSTEP, or BEST, analysis was used to test the correlation of environmental variables against differences in community composition among samples to determine which environmental variables most correlated to the differences in abundance among the community (Clarke et al. 2014). Environmental data were analyzed with a Draftsman plot to assess the skew and correlation between environmental variables. If the correlations between environmental variables were greater than 0.8, only one of the two covariables were included, and the data were normalized (Clarke et al. 2014). A resemblance matrix based on Euclidean distance was created for the normalized environmental data. Time series of significant environmental variables were plotted to assess change through time. Comparisons 1, 2, 3, and 4 employed BEST analysis (Table 3).

Abundance biomass comparison (ABC) curves were plotted for each year and station of both the macrofaunal and epifaunal communities for the current study. ABC curves plot cumulative dominance percentage biomass and cumulative dominance percentage abundance against ranked species to compare species dominance in abundance and biomass and to interpret the level of disturbance for each assemblage (Clarke 1990, Warwick & Clarke 1994) The W statistic reported for each plot is a measure of how far the biomass curve lies above the abundance curve. The closer to one the W statistic is, the higher above and more separated the biomass curve is from the abundance curve. Comparisons 1 and 2 employed ABC curves.

Taxonomic distinctness was used to compare current abundance data, including replicates, against all historical macrofaunal studies listed in Table 1 for all three stations. Taxonomic distinctness is a measure of biodiversity comparing taxonomic relatedness of species to one another within a sample (Clarke et al. 2014). It is independent of the number of species within a sample, allowing for comparison across varying methodologies (Clarke et al. 2014, Hale et al. 2018b). RStudio was used to plot box plots of the change in taxonomic distinctness through time for each station.

BIOMASS

Biomass was determined by first measuring wet weight of each species from each station. Large epifaunal wet weight was measured at sea.

Macrofaunal and small epifaunal wet weights were measured by blotting the animals dry and weighing them on a standard gram scale or microgram scale.

For some of the smaller and more abundant species, a subset was measured by size range of each species to calculate average individual wet weight, and abundance was used to calculate an approximate wet weight. Wet weights were converted to dry weight, shell-free dry weight, and then ash-free dry weight, the reported biomass measurement, using weight-to-weight conversion factors (Ricciardi & Bourget 1998).

RESULTS

ENVIRONMENTAL DATA

A gradient of salinity, temperature, dissolved oxygen, and sediment nitrogen and carbon content was evident along the West Passage of Narragansett Bay (Table 2). Bottom temperature decreased from the upper Bay PR station to the lower Bay RI station, salinity increased, dissolved oxygen increased and peaked at the NJ mid Bay station, decreasing south to the RI station, and both sediment carbon and nitrogen content decreased (Table 2).

Yearly average bottom dissolved inorganic nitrogen (DIN) decreased from 1975-2022, as did the individual components of both nitrate and nitrite ($\text{NO}_3 + \text{NO}_2$) and ammonium (NH_4) (Figure 2). Summer average surface temperature increased from 1975-2022. At both the NJ and PR stations, summer average bottom temperature, summer average bottom salinity, and summer average dissolved oxygen increased between 2000 and 2022 (Figure 3).

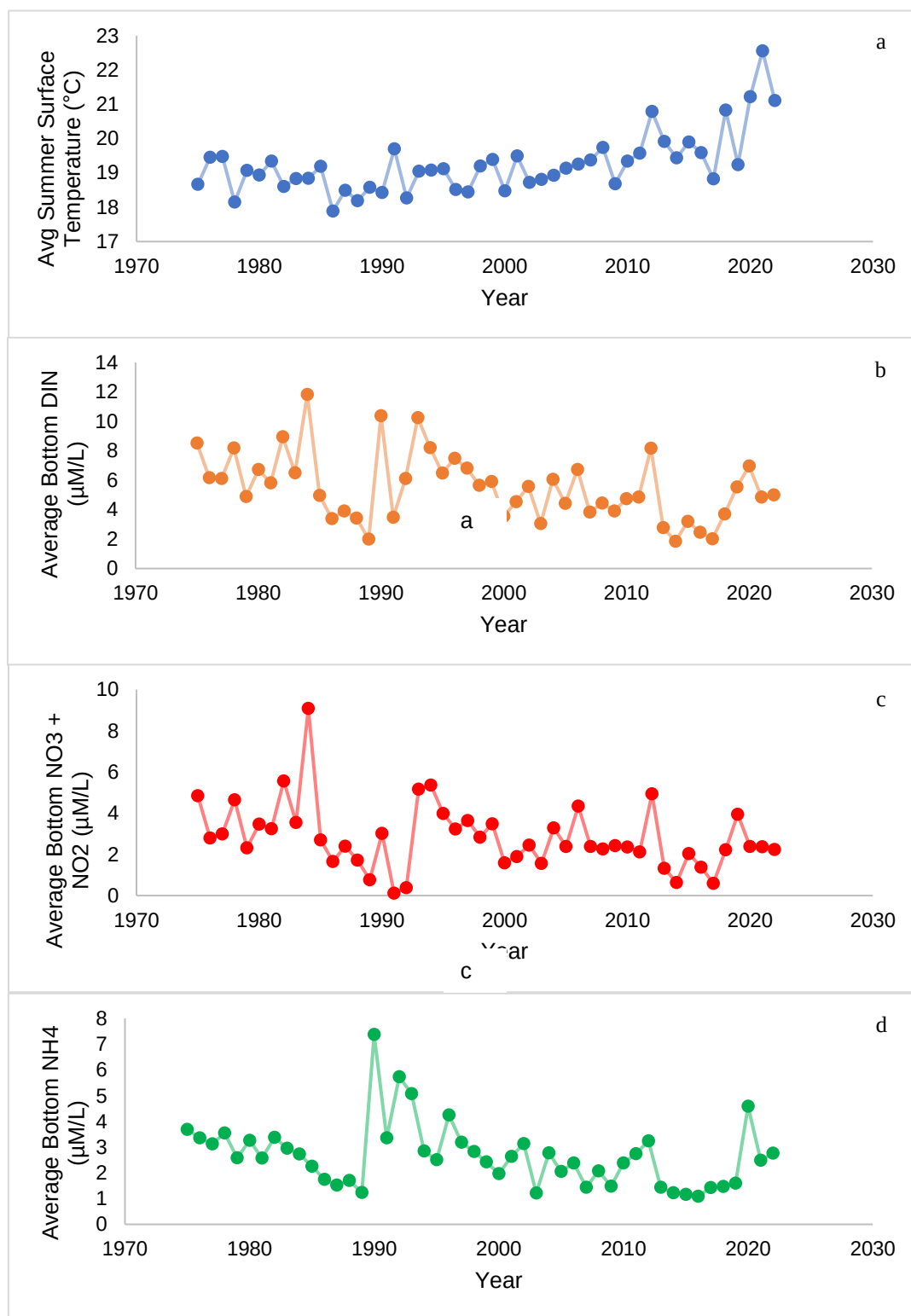


Figure 2: Environmental data for 1975-2022. Time series of relevant environmental variables for the comparison between the current core data and the historical core data. Locations for each dataset are cited in Table 3. a) temperature b) DIN c) NO₂+3 d) NH₄

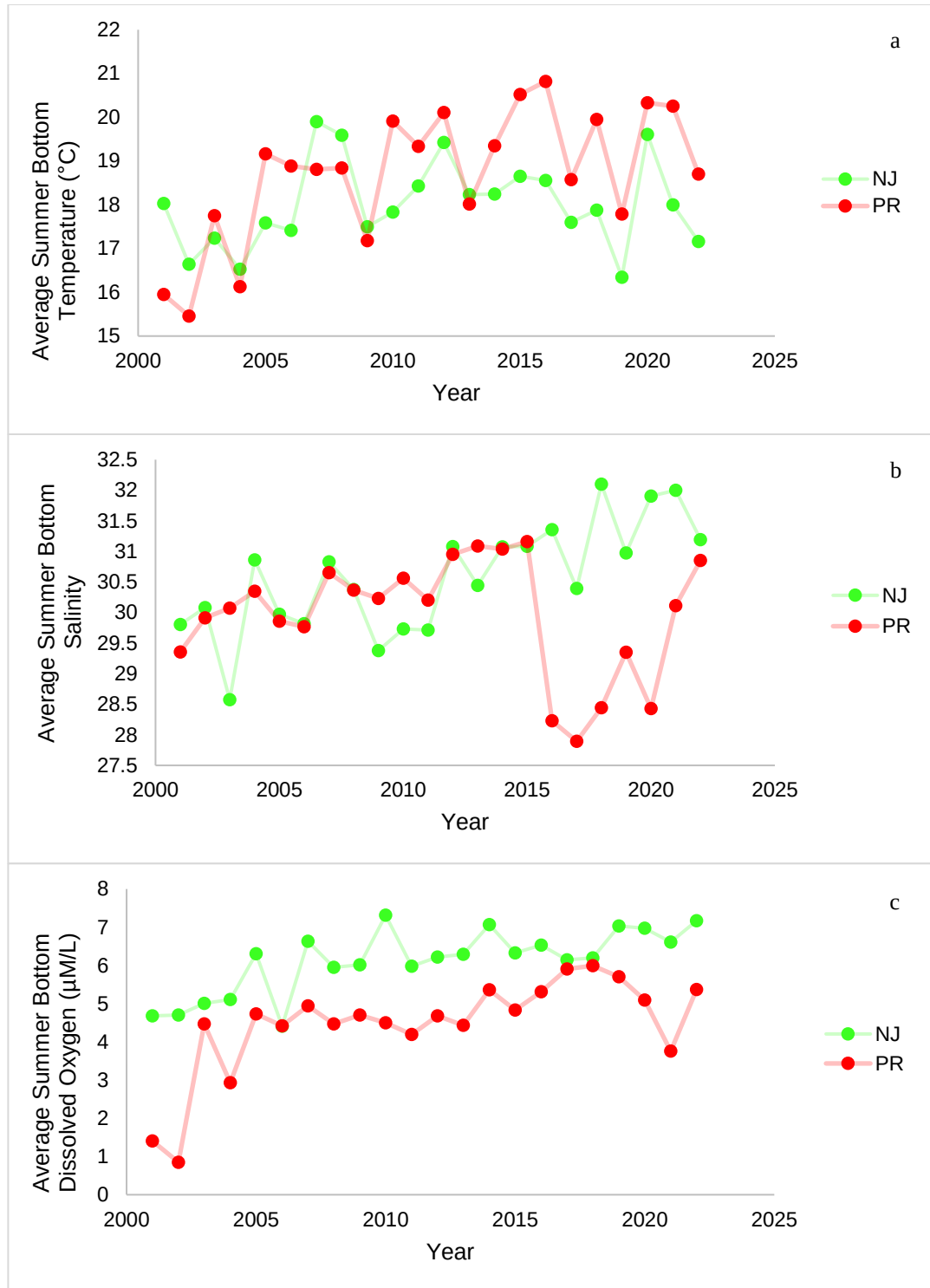


Figure 3: Environmental data for 2001-2022. Time series of relevant environmental variables for the comparison between the current core data and the core data from the study performed directly before the removal of tertiary nitrogen input from WWTF (Calabretta and Oviatt 2008). All data were averaged over the months of May to September of the corresponding year. a) temperature b) salinity c) dissolved oxygen

1: CURRENT MACROFAUNAL COMPARISON

Average total macrofaunal abundance was greater in 2021 for all stations than 2022. At the PR station, the 2022 average total abundance was almost half the 2021 abundance, with almost all replicate total abundances of 2022 less than the minimum replicate total abundance of 2021 (Figure 4a). NJ had the highest replicate total abundance in both 2021 and 2022. Average total biomass patterns mirrored those of abundance, greater in 2021 for all stations than 2022, again with the most marked decrease in replicate biomass between years occurring at the PR station. PR had the highest total biomass in 2021, but RI had the highest total biomass in 2022, while also possessing the lowest total abundance (Figure 4b). Total number of species also mirrored the pattern of total replicate abundance at the PR station; PR replicate total number of species decreased between sampling years (Figure 5a). The NJ and RI stations had fairly consistent replicate total species between years. PR had the highest replicate total species in 2021, while NJ had the highest in 2022. Pielou's index of species evenness increased among replicates between years at all three stations (Figure 5b). The RI stations had the highest species evenness in both years among replicates.

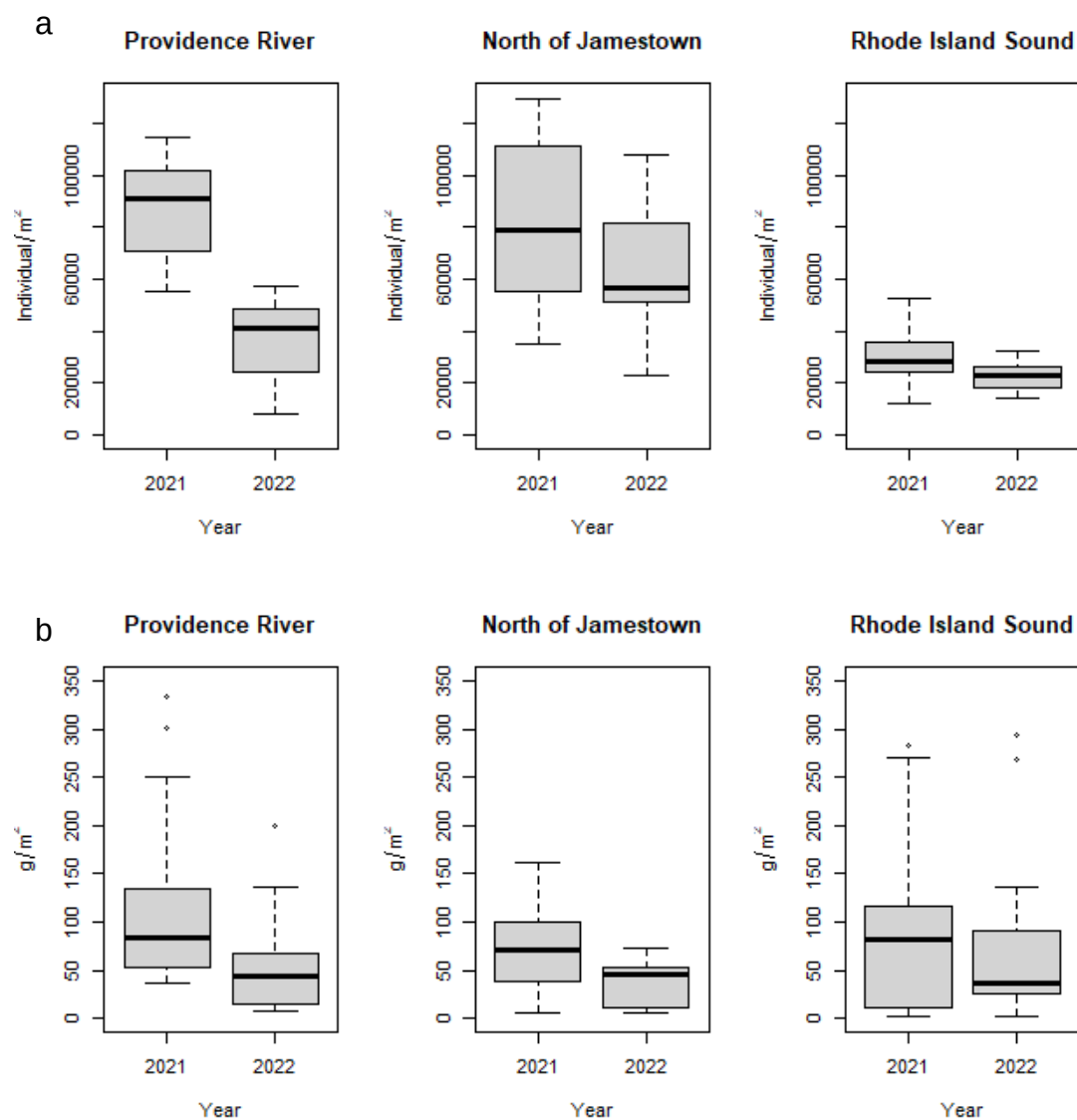


Figure 4: Macrofaunal biomass and abundance box plots displaying a) total abundance and b) total biomass at each station in 2021 and 2022 for the macrofaunal community of the current study, including replicates.

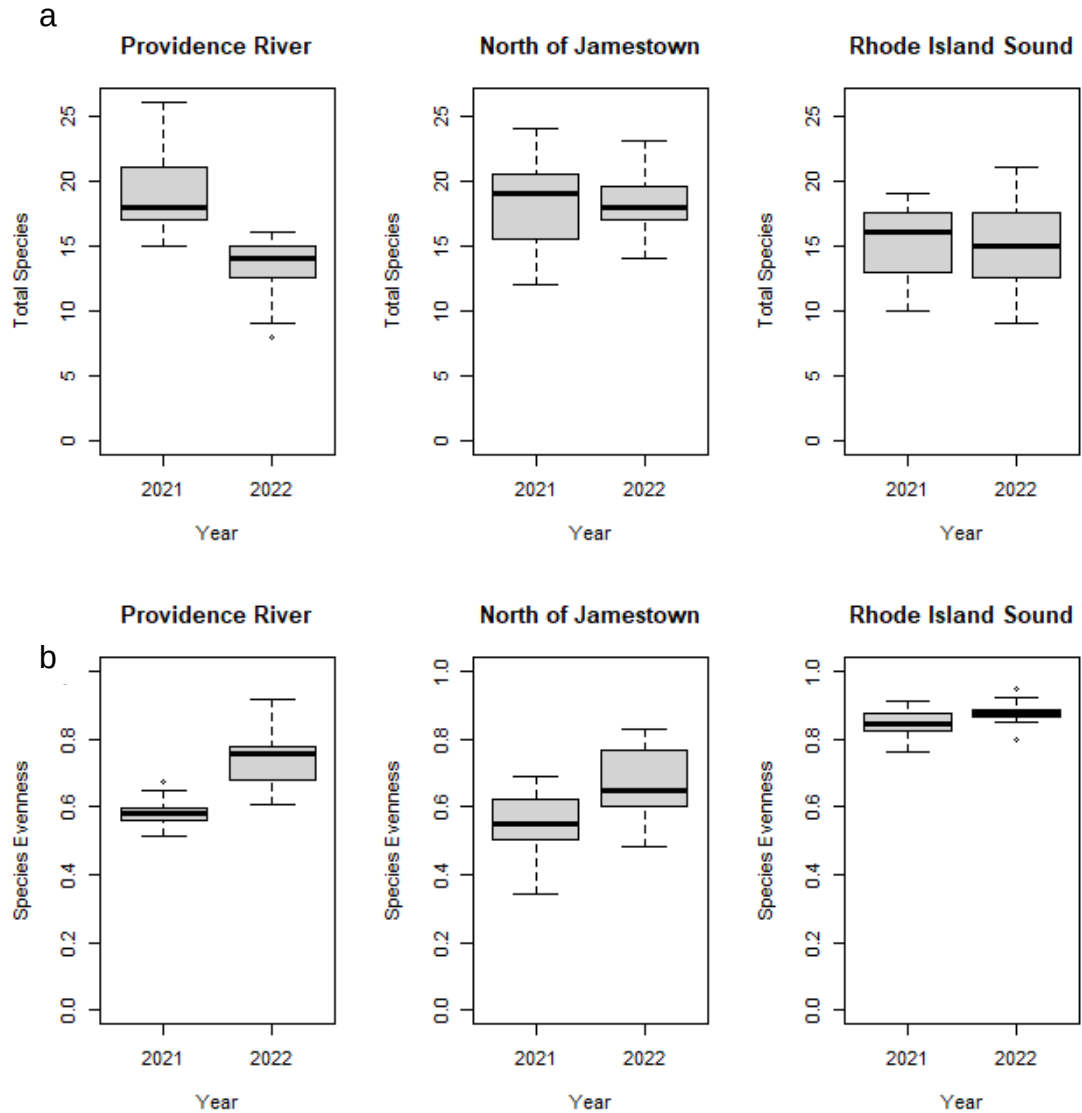


Figure 5: Macrofaunal total species and species evenness box plots displaying a) total number of species and b) Pielou's index of species evenness at each station in 2021 and 2022 for the macrofaunal community of the current study, including replicates.

Mediomastus ambiseta, a capitellid polychaete, was the most abundant species, in both years, for the PR station and the NJ station, and was among the top five for the RI station in both years (Table 4). The abundance of *M. ambiseta* decreased from 2021 to 2022 for all three stations, most notably at the PR station. *Streblospio benedicti*, a spionid polychaete, abundance also decreased between 2021 and 2022. *Ampelisca*, an amphipod, was the second most abundant species in 2021 at PR, but was not among the top five in 2022, replaced by a larger amphipod, *Unciola irrorata*, which resides in abandoned *Ampelisca* tubes (Table 4). *Nucula proxima* was the most abundant species in both years for the RI station, and was among the top five in both years for PR, and in 2021 for NJ. *N. proxima* also decreased in abundance at all three stations between 2021 and 2022.

Table 4: Top five macrofaunal species by abundance in individuals/m².

PR 21		NJ 21		RI 21	
Species	Average Abundance (#/m ²)	Species	Average Abundance (#/m ²)	Species	Average Abundance (#/m ²)
<i>Mediomastus ambiseta</i>	39,880	<i>Mediomastus ambiseta</i>	51,448	<i>Nucula proxima</i>	6,565
<i>Ampelisca</i>	24,057	<i>Nucula proxima</i>	4,218	<i>Paradoneis lyra</i>	3,265
<i>Nucula proxima</i>	5,986	<i>Levinsenia gracilis</i>	3,537	<i>Capitella capitata</i>	2,993
<i>Polydora cornuta</i>	2,041	<i>Scoletoma tenuis</i>	2,653	<i>Mediomastus ambiseta</i>	2,653
<i>Streblospio benedicti</i>	1,429	<i>Paradoneis lyra</i>	2,619	<i>Ninoe nigripes</i>	2,551
PR 22		NJ 22		RI 22	
Species	Average Abundance (#/m ²)	Species	Average Abundance (#/m ²)	Species	Average Abundance (#/m ²)
<i>Mediomastus ambiseta</i>	12,993	<i>Mediomastus ambiseta</i>	32,449	<i>Nucula proxima</i>	4,388
<i>Streblospio benedicti</i>	7,313	<i>Levinsenia gracilis</i>	5,476	<i>Paradoneis lyra</i>	2,857
<i>Oligochaeta</i>	4,898	<i>Paradoneis lyra</i>	4,082	<i>Mediomastus ambiseta</i>	1,871
<i>Nucula proxima</i>	2,313	<i>Chaetozone sp.</i>	2,483	<i>Chaetozone sp.</i>	1,531
<i>Unciola irrorata</i>	1,667	<i>Nemertea</i>	2,109	<i>Levinsenia gracilis</i>	1,531

A different assemblage dominated biomass (Table 5). *Nephtys incisa*, a large, bioturbating, nephtyid polychaete, had the second largest biomass in both years for the PR station, and the largest biomass in both years for the NJ and RI stations. *Mercenaria mercenaria*, the northern quahog, had the largest biomass for PR in 2021, also appearing in the top five biomasses of RI in both years. *Crangon septemspinosa*, a sand shrimp, had the largest biomass for PR in 2022, also appearing in the top five biomasses of NJ in 2022, and RI in both years (Table 5).

Table 5: Top five macrofaunal species by ash-free dry weight biomass in g/m².

PR 21		NJ 21		RI 21	
Species	Average Biomass (g/m ²)	Species	Average Biomass (g/m ²)	Species	Average Biomass (g/m ²)
<i>Mercenaria mercenaria</i>	38.8	<i>Nephtys incisa</i>	51.7	<i>Nephtys incisa</i>	39.6
<i>Nephtys incisa</i>	22.4	<i>Anomia simplex</i>	6.9	<i>Mercenaria mercenaria</i>	25.9
<i>Crepidula fornicata</i>	18.6	<i>Mediomastus ambiseta</i>	3.3	<i>Havelockia scabra</i>	7.9
<i>Crangon septemspinosa</i>	15.0	<i>Ilyanassa trivittata</i>	1.7	<i>Crangon septemspinosa</i>	2.2
<i>Cistenides granulata</i>	8.5	<i>Glycera dibranchiata</i>	1.6	<i>Ilyanassa trivittata</i>	2.1
PR 22		NJ 22		RI 22	
Species	Average Biomass (g/m ²)	Species	Average Biomass (g/m ²)	Species	Average Biomass (g/m ²)
<i>Crangon septemspinosa</i>	21.8	<i>Nephtys incisa</i>	19.0	<i>Nephtys incisa</i>	36.2
<i>Nephtys incisa</i>	10.3	<i>Crangon septemspinosa</i>	4.6	<i>Mercenaria mercenaria</i>	25.9
<i>Cistenides granulata</i>	5.8	<i>Cistenides granulata</i>	2.2	<i>Crangon septemspinosa</i>	4.6
<i>Macoploma tenta</i>	4.7	<i>Mediomastus ambiseta</i>	2.1	<i>Havelockia scabra</i>	3.9
<i>Ilyanassa trivittata</i>	3.9	<i>Ilyanassa trivittata</i>	1.7	<i>Glycera dibranchiata</i>	1.7

ABC curves showed evidence for less disturbed communities for all stations and years (Figure 6). All biomass curves were consistently above abundance curves, with $W > 0.6$, indicating macrofaunal benthic communities at all stations in all years contained more large-bodied bioturbators dominating percent cumulative biomass with low percent cumulative abundance (Figure 6). This pattern indicated higher successional stage benthic communities at all three stations sampled. This result is supported by the dominating biomass of *Nephtys incisa* at all three stations, as well as large-bodied bivalves (Table 5). As benthic successional stage increases, small, highly abundant, opportunistic, surface-dwelling macrofauna are replaced with larger, less abundant, deeper bioturbators as the layer of oxygenated sediment becomes deeper with less eutrophic disturbance.

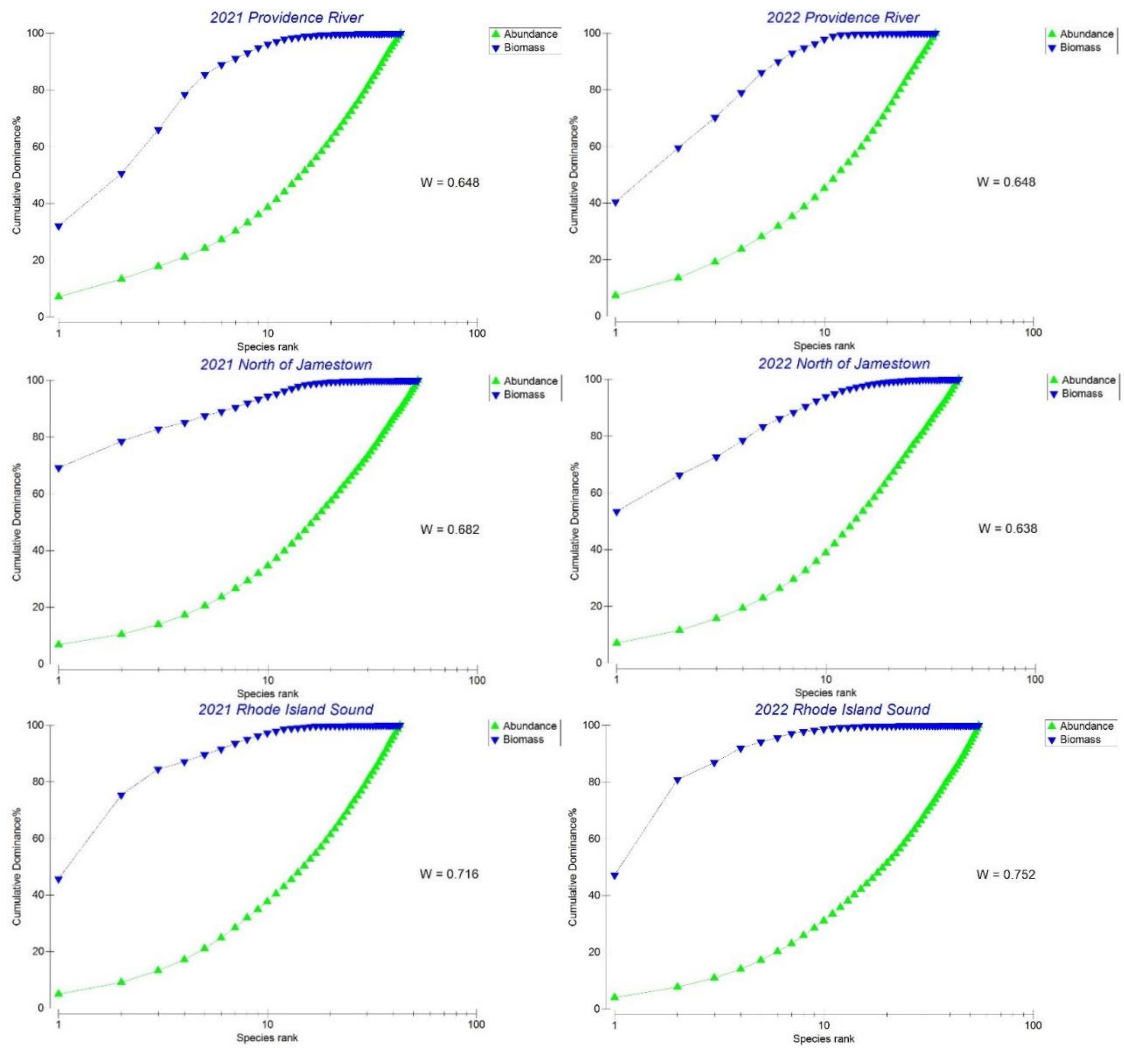


Figure 6: Macrofaunal abundance biomass curves. Abundance to biomass comparison curves for the macrofaunal benthic community averaged over core replicates for each station. The W statistic is a measure of how high above the abundance curve the biomass curve lies.

The MDS analyses for macrofauna revealed spatial distinctions of a clear separation between each station, from Providence River to North of Jamestown to Rhode Island Sound (Figure 7). The 2021 Providence River data clustered separately from the 2022 Providence River data. Rhode Island Sound and North of Jamestown clusters show no clear separation between years. Aside from one North of Jamestown and two Rhode Island Sound replicate outliers, replicate samples from each station formed tight clusters, indicating similar species composition among replicates. The clear separation between stations suggests that the change from Providence River to North of Jamestown to Rhode Island Sound reflects the environmental gradient down the West Passage of the Bay (Table 2). The visual MDS separation of stations is supported by the ANOSIM performed on macrofauna (Table 6). The global R statistic was greater than 0.8 with $p = 0.001$, indicating significant dissimilarities between stations. Pairwise station comparisons for macrofaunal communities were significant, with Rhode Island Sound and Providence River being the most dissimilar, and Rhode Island Sound and North of Jamestown being the most similar (Table 6).

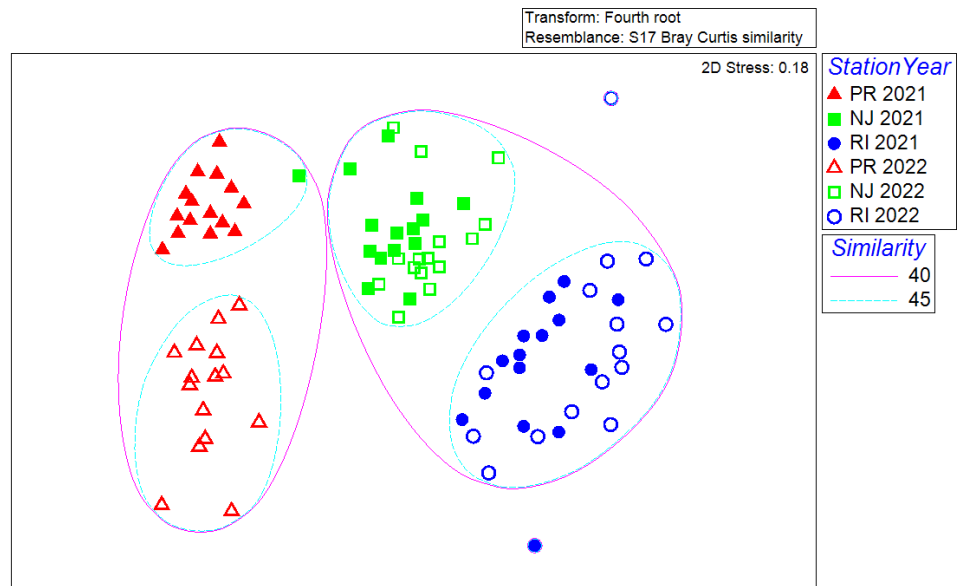


Figure 7: Comparison 1 multi-dimensional scaling plot, displaying the between station and between year shifts in Bray-Curtis dissimilarity for the current core data: PR: Providence River, NJ: North of Jamestown, RI: Rhode Island Sound. Axes do not have units; it is a way to visualize the relative similarities and differences in species composition between each replicate sample in two dimensions. Colored ellipses of 45% and 40% surrounding clusters indicate percent similarity within those clusters.

Table 6: Comparison 1 analysis results. Statistical analysis of current macrofaunal community composition. The test statistic R measures the difference between the average of rank similarities among replicates *within* sites and the average of rank similarities from pairs of replicates *between* sites. R=1 if *all* replicates within sites are more similar to each other than *any* replicates from different sites. The p-value is the significance level (Clarke 1993).

Test	Group	R	p
ANOSIM	Global	0.883	0.001
	PR vs NJ	0.943	0.001
	PR vs RI	0.986	0.001
	NJ vs RI	0.743	0.001
SIMPER	Species	PR average abundance	NJ average abundance
PR vs NJ	<i>Levinsenia gracilis</i>	0	6.86
	<i>Streblospio benedicti</i>	6.75	0.16
	<i>Paradoneis lyra</i>	0	6.55
	<i>Ampelisca</i>	7.55	3.06
	<i>Clymenella torquata</i>	0.85	4.61
SIMPER	Species	PR average abundance	RI average abundance
PR vs RI	<i>Paradoneis lyra</i>	0	7.16
	<i>Streblospio benedicti</i>	6.75	0.16
	<i>Ampelisca</i>	7.55	0.66
	<i>Mediomastus ambiseta</i>	12.1	6.18
	<i>Levinsenia gracilis</i>	0	5.46
SIMPER	Species	NJ average abundance	RI average abundance
NJ vs RI	<i>Mediomastus ambiseta</i>	13.78	6.18
	<i>Ninoe nigripes</i>	0.38	5.36
	<i>Polydora cornuta</i>	5.06	0.32
	<i>Spiochaetopterus oculatus</i>	5.12	0.66
	<i>Nemertea</i>	5.24	2.25
SIMPER	Species	PR 2021 average abundance	PR 2022 average abundance
PR 2021 vs 2022	<i>Ampelisca</i>	12.19	2.9
	<i>Oligochaeta</i>	0	8.15
	<i>Acteocina canaliculata</i>	4.72	0
	<i>Turbonilla interrupta</i>	5.31	0.95
	<i>Streblospio benedicti</i>	4.71	8.79
BEST	Stressor variables		Rho
	Dissolved oxygen	Temperature	0.9
	Sediment percent carbon	Sediment percent nitrogen	

SIMPER identified the species contributing the highest percentage differences between station communities (Table 6). Certain species, such as *Streblospio benedicti*, a tube-dwelling spionid polychaete, is quite abundant in the Providence River community, but much less abundant in both the North of Jamestown and the Rhode Island Sound communities (Table 4, Table 6). *Ampelisca*, tube-dwelling amphipods, follow a similar trend. Both *S. benedicti* and *Ampelisca* are tube-dwelling, suspension feeding opportunists. In contrast, the paraonid polychaetes *Levinsenia gracilis* and *Paradoneis lyra*, are both abundant in North of Jamestown and Rhode Island Sound, but absent from the Providence River assemblage (Table 4, Table 6). Paraonid polychaetes are deposit feeders that either do not prefer, or cannot tolerate the temperature, salinity, dissolved oxygen, or organic matter content of the upper Bay. Concentrated mats of tube-dwellers may also change the biochemistry of sediment and prevent the settlement of deposit feeders (NBNERR 2009).

The separation between the macrofaunal communities of the Providence River station in 2021 and 2022 in the MDS plot is explained by the marked decrease in abundance between years, and is further shown by the SIMPER analysis performed between years (Figure 4, Figure 7, Table 6). *Ampelisca* was much more abundant in 2021 than 2022, suggesting divers collecting samples in 2021 encountered an aggregation of *Ampelisca*, which tend to form dense mats of tubes, especially above hypoxic, soft sediment. Oligochaetes were completely absent from the 2021 Providence River assemblage, while being quite abundant in the 2022 assemblage, whereas

Acteocina canaliculata, a bubble shell gastropod very sensitive to disturbance, had the opposite trend (Table 6). Both *Ampelisca* amphipods and oligochaetes are second-order opportunists with the ability to tolerate high organic matter and low oxygen habitats, but also have short lifespans. The more sensitive *Acteocina canaliculata* may have found suitable settlement habitat in 2021, but either settlement did not take in 2022, or the sensitive organisms died from a stressor after sampling and before reproducing.

The environmental variables best correlating to the changes in community composition between macrofaunal stations and years were temperature, dissolved oxygen concentration, sediment nitrogen, and sediment carbon at ρ (rho) = 0.9 (Table 6). The results of the BEST analysis further support the environmental gradient-driven differences in community composition at each station. Significant dissimilarities between stations suggest unique communities at each station, driven by the gradients of decreasing organic matter, reflected by sediment and carbon nitrogen content, decreasing bottom temperature, and increasing bottom water dissolved oxygen (Table 2). Epifaunal biomass was also tested, but was not among the most correlated environmental variables.

2: CURRENT EPIFAUNAL COMPARISON

Total replicate epifaunal abundance was greater at the NJ and RI stations in 2022 compared to 2021 (Figure 8a). The PR station had the highest total replicate abundance in 2021, but NJ 2022 had a much higher replicate abundance than any other station. Total replicate biomass for all stations decreased between 2021 and 2022 (Figure 8b). The PR station had the greatest total replicate biomass in 2021, and the greatest decrease in replicate total biomass between sampling years, while total abundance stayed consistent year-to-year (Figure 7). Replicate total number of species was consistent year to year for the PR station, while it increased for the NJ and RI station (Figure 9a). The PR station had the highest replicate total number of species in 2021, while the NJ station had the highest in 2022. Pielou's index of species evenness decreased at the NJ station between years, while increasing at the PR and RI stations between years (Figure 9b). The NJ station had the highest replicate species evenness in 2021, and the PR station had the highest replicate species evenness in 2022.

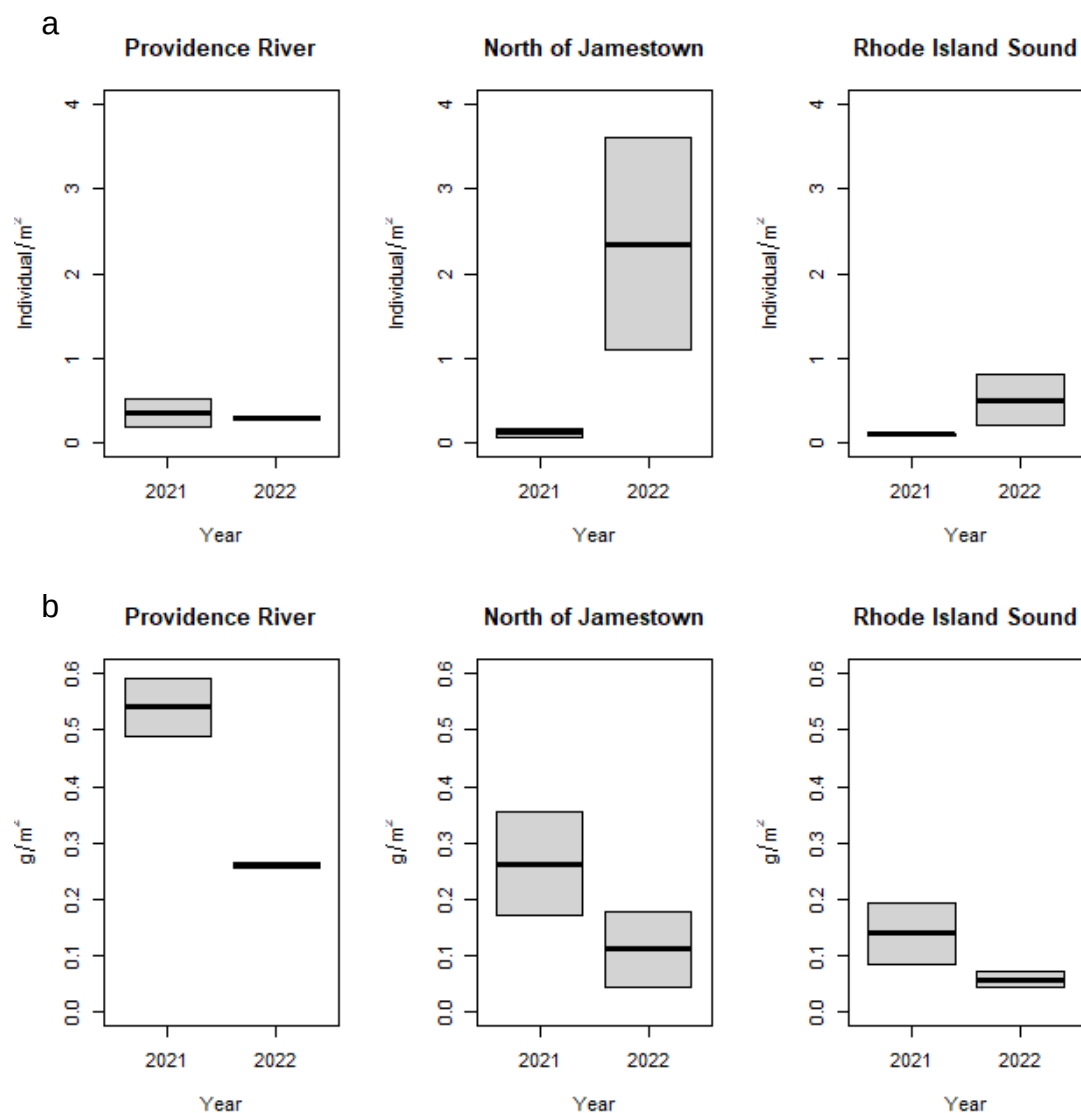


Figure 8: Epifaunal biomass and abundance box plots displaying a) total abundance and b) total biomass at each station in 2021 and 2022 for the macrofaunal community of the current study, including replicates.

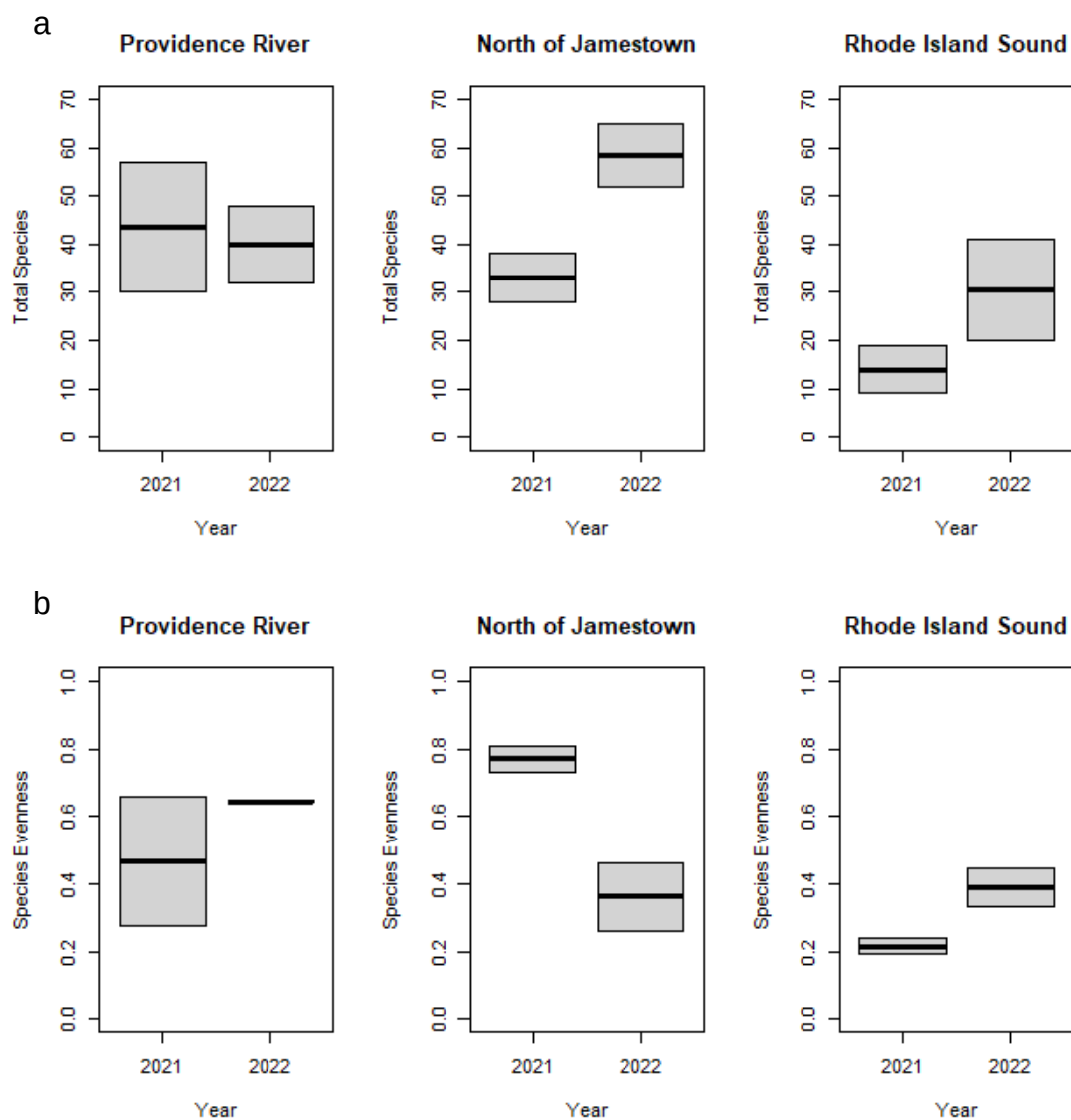


Figure 9: Epifaunal total species and species evenness box plots displaying a) total number of species and b) Pielou's index of species evenness at each station in 2021 and 2022 for the macrofaunal community of the current study, including replicates.

Polydora cornuta was the most abundant epifaunal species caught for PR in both 2021 and 2022, and for NJ and RI in 2022 (Table 7). *Crepidula fornicata*, the common slipper shell, was also among the top five most abundant species for all three sampling stations in 2022. *P. cornuta* is often found in the empty shells of *C. fornicata*, suggesting an aggregation from the collection of a *Crepidula* reef in the trawl at all three stations, especially NJ, in 2022 (Table 7). The tubes of *U. irrorata* were probably also dragged up and caught in the net, despite their small size, with large mats of tubes, bacterial mats, algae, and sediment at the PR station in both years and the NJ station in 2022 (Table 7). *Crangon septemspinosa* was the most abundant species for RI in 2021, and the second most abundant species in 2022.

Table 7: Top five epifaunal species by abundance in individuals/m².

PR 21		NJ 21		RI 21	
Species	Average Abundance (#/m ²)	Species	Average Abundance (#/m ²)	Species	Average Abundance (#/m ²)
<i>Polydora cornuta</i>	0.101	<i>Ilyanassa trivittata</i>	0.033	<i>Crangon septemspinosa</i>	0.087
<i>Ampelisca abdita</i>	0.085	<i>Busycotypus canaliculatus</i>	0.011	<i>Cancer irroratus</i>	0.003
<i>Clymenella torquata</i>	0.027	<i>Mytilus edulis</i>	0.007	<i>Urophycis regia</i>	0.0007
<i>Harmothoe extenuata</i>	0.016	<i>Macoploma tenta</i>	0.006	<i>Cancer borealis</i>	0.0006
<i>Unciola irrorata</i>	0.015	<i>Pagurus longicarpus</i>	0.005	<i>Havelockia scabra</i>	0.0006
PR 22		NJ 22		RI 22	
Species	Average Abundance (#/m ²)	Species	Average Abundance (#/m ²)	Species	Average Abundance (#/m ²)
<i>Polydora cornuta</i>	0.103	<i>Polydora cornuta</i>	1.676	<i>Polydora cornuta</i>	0.228
<i>Edwardsia elegans</i>	0.038	<i>Crepidula fornicata</i>	0.141	<i>Crangon septemspinosa</i>	0.143
<i>Unciola irrorata</i>	0.035	<i>Unciola irrorata</i>	0.11	<i>Clymenella torquata</i>	0.06
<i>Crepidula fornicata</i>	0.026	<i>Ilyanassa trivittata</i>	0.109	<i>Mediomastus ambiseta</i>	0.017
<i>Libinia emarginata</i>	0.012	<i>Clymenella torquata</i>	0.068	<i>Crepidula fornicata</i>	0.012

The top five epifaunal species by biomass revealed a more typical mobile predator assemblage (Table 8). *Libinia emarginata*, the common spider crab, had the largest biomass for PR in 2022 and NJ in 2021, as well as the second largest biomass for PR in 2021 and NJ in 2022. *Callinectes sapidus*, the blue crab and *Mercenaria mercenaria*, the northern quahog, were in the top five for PR both years. *Pagurus pollicaris*, the flat-clawed hermit crab, had the second largest biomass for NJ in 2021, and the first largest biomass for NJ in 2022. *Cancer irroratus*, the Atlantic rock crab, had the largest biomass for RI in both years (Table 8). Despite its small size, *Crangon septemspinosa* had the fifth largest biomass for RI in both years, probably due to the high abundance (Table 7).

Table 8: Top five epifaunal species by ash-free dry weight biomass in g/m².

PR 21		NJ 21		RI 21	
Species	Average Biomass (g/m ²)	Species	Average Biomass (g/m ²)	Species	Average Biomass (g/m ²)
<i>Limulus polyphemus</i>	0.236	<i>Libinia emarginata</i>	0.062	<i>Cancer irroratus</i>	0.083
<i>Libinia emarginata</i>	0.219	<i>Pagurus pollicaris</i>	0.049	<i>Leucoraja erinacea</i>	0.034
<i>Callinectes sapidus</i>	0.030	<i>Busycotypus canaliculatus</i>	0.044	<i>Cancer borealis</i>	0.010
<i>Busycotypus canaliculatus</i>	0.018	<i>Cliona celata</i>	0.038	<i>Hippoglossina oblonga</i>	0.005
<i>Mercenaria mercenaria</i>	0.014	<i>Paralichthys dentatus</i>	0.032	<i>Crangon septemspinosa</i>	0.003
PR 22		NJ 22		RI 22	
Species	Average Biomass (g/m ²)	Species	Average Biomass (g/m ²)	Species	Average Biomass (g/m ²)
<i>Libinia emarginata</i>	0.217	<i>Pagurus pollicaris</i>	0.028	<i>Cancer irroratus</i>	0.019
<i>Pseudopleuronectes americanus</i>	0.024	<i>Libinia emarginata</i>	0.018	<i>Homarus americanus</i>	0.012
<i>Clathria (Clathria) prolifera</i>	0.005	<i>Busycon carica</i>	0.017	<i>Etropus microstomus</i>	0.008
<i>Callinectes sapidus</i>	0.004	<i>Paralichthys dentatus</i>	0.016	<i>Merluccius bilinearis</i>	0.007
<i>Mercenaria mercenaria</i>	0.003	<i>Prionotus evolans</i>	0.009	<i>Crangon septemspinosa</i>	0.005

Epifaunal benthic communities at all stations in both years also presented as less disturbed, according to the ABC curves (Figure 8). Mirroring the macrofaunal ABC curves, the biomass curve was above the abundance curve for all plots, $W > 0.6$, indicating that large-bodied, low abundance organisms dominated biomass, especially for PR in both years. PR in both years had a larger gap between the abundance and biomass curves than either NJ or RI, suggesting the beam trawl tows at PR for both years caught more large-bodied, low abundance epifauna, including large bivalves, large decapods, and large fish (Table 7, 8). The NJ and the RI beam trawl tows may have dragged up a higher abundance of smaller organisms caught in algae and sediment tube and bacterial mats.

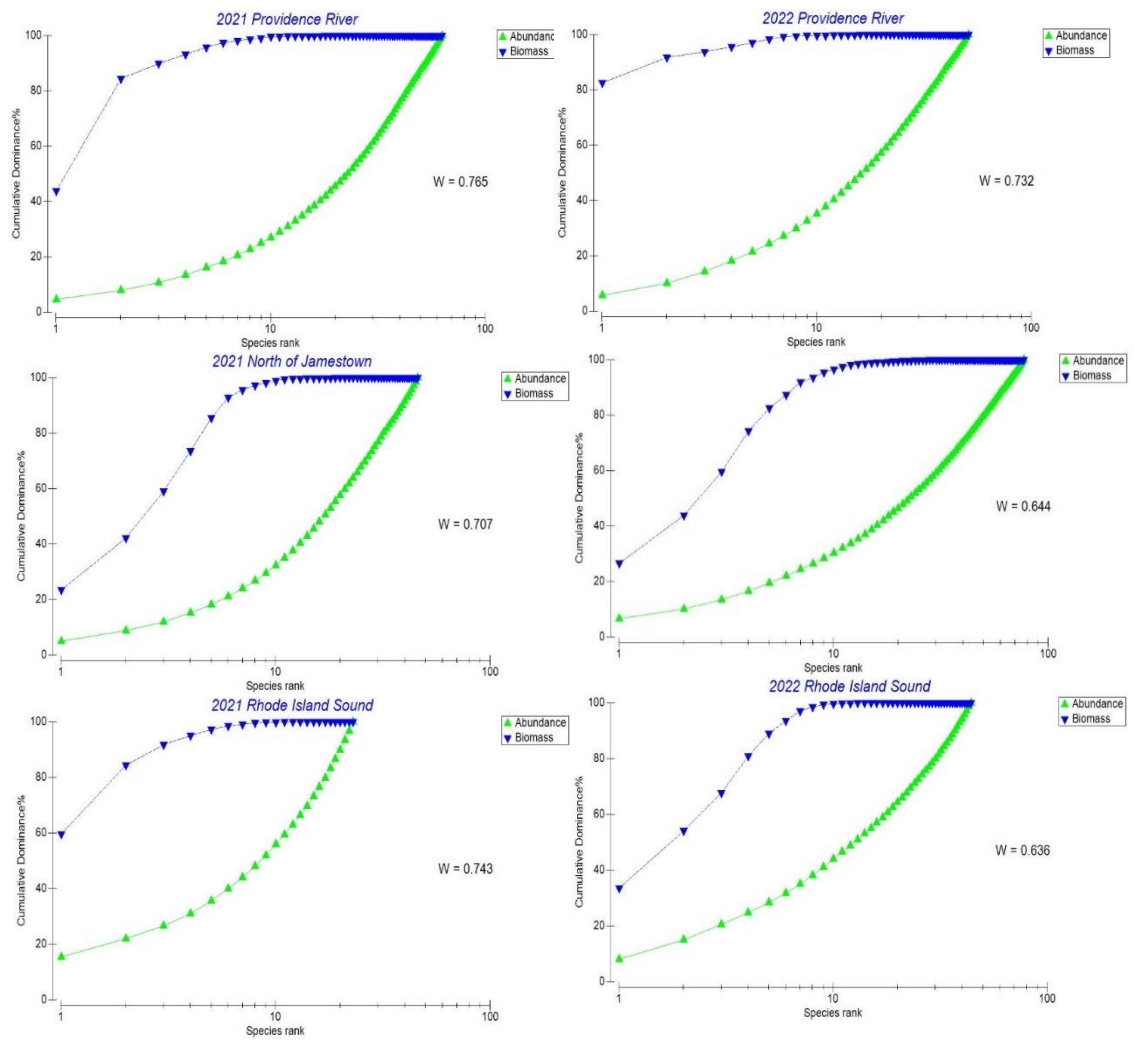


Figure 10: Epifaunal abundance biomass curves. Abundance to biomass comparison curves for the epifaunal benthic community averaged over beam trawl replicates for each station. The W statistic is a measure of how high above the abundance curve the biomass curve lies.

Whereas all other replicates are within 50% similarity clusters of one another, the replicates for Providence River 2021 have quite a separation between them (Figure 11). The replicates for both years for North of Jamestown and Providence River seem to cluster in a line apart from the Rhode Island Sound data, whose replicates for each year are also quite separated from one another (Figure 11). The difference in PR 2021 replicate assemblage is probably due to limited container space for the large amount of trawl material sampled in both replicates. The year-to-year difference in the RI assemblage is probably due to the deep depth of the site and the discrepancy of winch wire length between years.

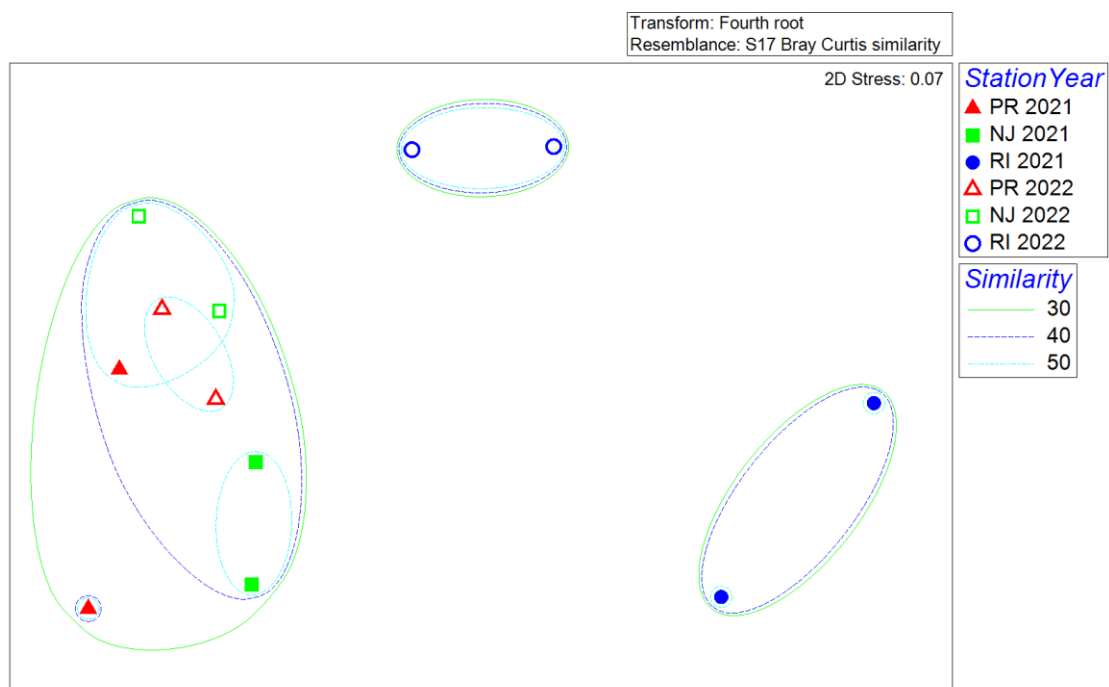


Figure 11: Comparison 2 multi-dimensional scaling plot, displaying the between station and between year shifts in Bray-Curtis dissimilarity for the current core data: PR: Providence River, NJ: North of Jamestown, RI: Rhode Island Sound. Axes do not have units; it is a way to visualize the relative similarities and differences in species composition between each replicate sample in two dimensions. Colored ellipses surrounding clusters indicate percent similarity within those clusters.

ANOSIM revealed a significant global statistic between stations of $R = 0.861$, $p = 0.004$, suggesting significantly different epifaunal communities down the gradient of the West Passage (Table 9). The pairwise statistics for direct comparisons between Providence River station and Rhode Island Sound and between North of Jamestown and Rhode Island Sound were reported as $R = 1$ for both, suggesting complete dissimilarity between Rhode Island Sound and the other two stations. Due to the small number of replicates and possible permutations of ANOSIM, however, the R values were reported at $p = 0.111$ (Table 9).

Table 9: Comparison 2 analysis results. Statistical analysis of current epifaunal community composition. The test statistic R measures the difference between the average of rank similarities among replicates *within* sites and the average of rank similarities from pairs of replicates *between* sites. R=1 if *all* replicates within sites are more similar to each other than *any* replicates from different sites. The p-value is the significance level (Clarke 1993).

Test	Group	R	p
ANOSIM	Global	0.861	0.004
	PR vs NJ	0.75	0.111
	PR vs RI	1	0.111
	NJ vs RI	1	0.111
SIMPER	Species	PR average abundance	NJ average abundance
PR vs NJ	<i>Polydora cornuta</i>	0.49	0.61
	<i>Ampelisca</i>	0.37	0.12
	<i>Clymenella torquata</i>	0.19	0.3
	<i>Ilyanassa trivittata</i>	0.27	0.49
	<i>Callipallene brevisrostris</i>	0.05	0.24
SIMPER	Species	PR average abundance	RI average abundance
PR vs RI	<i>Crangon septemspinosa</i>	0.2	0.58
	<i>Ampelisca</i>	0.37	0.04
	<i>Polydora cornuta</i>	0.49	0.31
	<i>Harmothoe extenuata</i>	0.3	0.04
	<i>Libinia emarginata</i>	0.29	0.04
SIMPER	Species	NJ average abundance	RI average abundance
NJ vs RI	<i>Ilyanassa trivittata</i>	0.49	0.08
	<i>Crangon septemspinosa</i>	0.26	0.58
	<i>Harmothoe extenuata</i>	0.34	0.04
	<i>Mytilus edulis</i>	0.19	0.11
	<i>Pagurus pollicaris</i>	0.23	0
BEST	Stressor variables	Rho	
	Temperature	0.789	
	Salinity		

SIMPER identified the species contributing the highest percentage differences between station communities (Table 9). Within the epifaunal community, *Crangon septemspinosa* was much more abundant in Rhode Island Sound than in either Providence River or North of Jamestown (Table 9). *Polydora cornuta*, a common spionid polychaete to the Bay, was much more abundant in North of Jamestown than either other station, although this high abundance may be explained by the second beam trawl replicate tow in North of Jamestown 2022 picking up an ephemeral reef of *Crepidula fornicata* shells, in which *P. cornuta* makes intensive use of the gastropod shell (Table 7, Table 9).

The environmental variables best correlating to the differences in community composition for epifauna are temperature and salinity at ρ (rho) = 0.789, reflective of the environmental gradient present down the West Passage (Table 9). Temperature decreases down the Bay, while salinity increases (Table 2). The changes in epifaunal community composition down the Bay are driven by temperature and salinity thresholds of organisms. Organisms preferring colder, saltier water will be found down the Bay near Rhode Island Sound, while organisms preferring warmer, fresher water will be found in the upper Bay, near Providence River.

3: BEFORE/AFTER TERTIARY NITROGEN MACROFAUNAL COMPARISON

The macrofaunal communities at the North of Jamestown station and the Providence River station were compared between the current study and

the study performed directly before the mandated removal of tertiary nitrogen input from WWTF (Calabretta and Oviatt 2008, referred to as 'Calabretta') (Figure 12, Table 10). The Providence River community composition had a large shift through time, as the Calabretta data from 2000-2004 Providence River clustered separately from the current 2021-2022 Providence River communities (Figure 12). The current Providence River communities, in fact, clustered more with the North of Jamestown data, suggesting that the Providence River macrofaunal community became more similar to the North of Jamestown community through time. There also appeared to be a separate clustering of 2000-2002 North of Jamestown communities from the later 2006-2010 communities, and even further from the current study 2021 and 2022 North of Jamestown communities, probably reflecting year-to-year variation within 50% similarity (Figure 12).

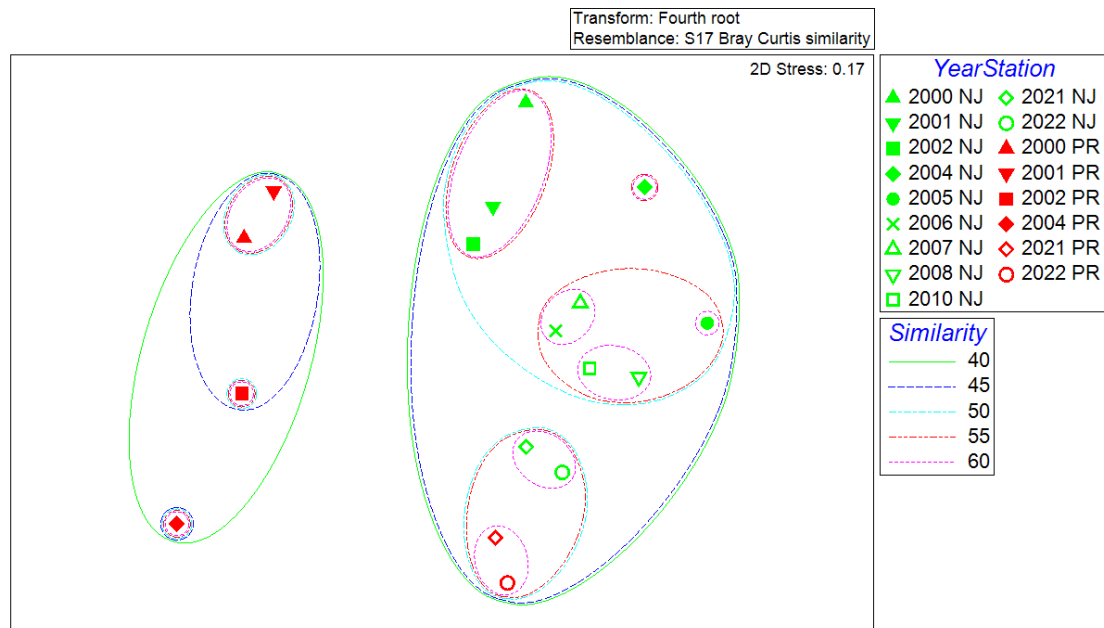


Figure 12: Comparison 3 multi-dimensional scaling plot, displaying the between year shifts in Bray-Curtis dissimilarity for the current core data compared to data from the study performed directly before the removal of tertiary nitrogen from WWTF (Calabretta and Oviatt 2008). Abundance data is aggregated to the taxonomic family level to decrease identification bias: PR: Providence River, NJ: North of Jamestown. Axes do not have units; it is a way to visualize the relative similarities and differences in species composition between each sample in two dimensions. Colored ellipses surrounding clusters indicate percent similarity within those clusters.

Table 10: Comparison 3 analysis results. Statistical analysis comparing current study macrofauna to the most recent historical study (Calabretta 2008). The test statistic R measures the difference between the average of rank similarities among replicates *within* sites and the average of rank similarities from pairs of replicates *between* sites. R=1 if *all* replicates within sites are more similar to each other than *any* replicates from different sites. The p-value is the significance level (Clarke 1993).

Test	Group	R	p
ANOSIM	Between studies	0.72	0.001
	Between stations	0.562	0.001
SIMPER	Species	Average abundance before	Average abundance after
	Between studies		
	Capitellidae	13.45	15.21
	Spionidae	9.11	8.44
	Nuculidae	5.42	7.16
	Paraonidae	1.74	6.71
	Cirratulidae	3.74	3.71
SIMPER	Species	PR average abundance	NJ average abundance
	Between stations		
	Spionidae	14.43	3.4
	Nuculidae	3.05	8.91
	Capitellidae	13.84	14.23
	Cirratulidae	3.85	3.62
	Ampeliscidae	5.79	2.73
BEST	Stressor variables	Rho	
	NO3 + NO2	0.605	
	Dissolved oxygen		
BEST	Stressor variables	Rho	
	NO3 + NO2	0.597	
	Temperature		
	Dissolved oxygen		

ANOSIM compared the Calabretta ‘before’ communities to the current ‘after’ communities, finding a significant difference between groups, $R = 0.72$, $p = 0.001$ (Table 10). SIMPER revealed the five taxonomic families contributing the most to differences between the before and after groups, capitellidae, spionidae, nuculidae, paraonidae, and cirratulidae, all polychaete worm families except nuculidae, a deposit-feeding bivalve family (Table 10). The environmental variables best correlating to the differences in community composition between the before and after groups were dissolved oxygen and nitrate + nitrite at $\rho = 0.605$. The second-best combination of environmental variables was temperature, dissolved oxygen and nitrate + nitrite at $\rho = 0.597$ (Table 10). Chlorophyll a was also tested, but was not among the most correlated environmental variables.

4: HISTORICAL CORE MACROFAUNAL COMPARISON

All historical core studies were then compared (Figure 13). The 1970s Providence River community began on the left side of the MDS plot, shifted to the right side in the early 2000s, and ended in the middle nearer to the North of Jamestown cluster in the 2020s. North of Jamestown communities of the mid-1970s clustered more with the 1976 Providence River community, but the succeeding North of Jamestown communities clustered more tightly in the middle of the plot (Figure 13). This mirrors the previous MDS comparing before and after tertiary nitrogen removal, which found the Providence River

community became clustered closer together to the North of Jamestown community through time (Figure 12).

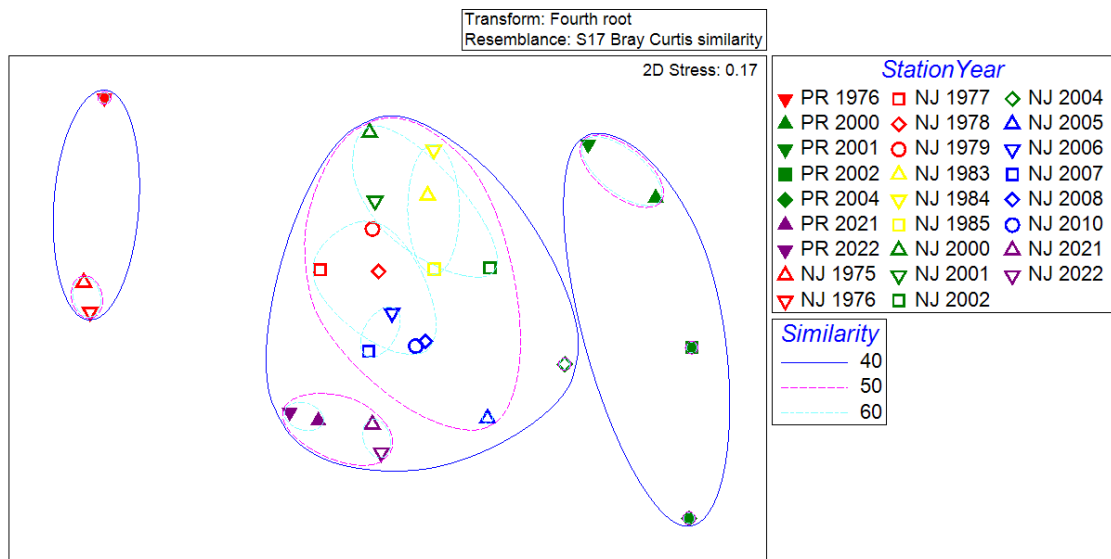


Figure 13: Comparison 4 multi-dimensional scaling plot, displaying the between year shifts in Bray-Curtis dissimilarity for the current core data compared to historical core data: PR: Providence River, NJ: North of Jamestown. Axes do not have units; it is a way to visualize the relative similarities and differences in species composition between each replicate sample in two dimensions. Colored ellipses surrounding clusters indicate percent similarity within those clusters. Color coded by decade, with open shapes representing North of Jamestown communities through time and solid shapes representing Providence River communities through time.

ANOSIM revealed a significant dissimilarity between years of macrofaunal communities through time, $R = 0.892$, $p = 0.001$ (Table 11). ANOSIM between stations was also analyzed, $R = 0.534$, $p = 0.001$, and SIMPER revealed the five taxonomic families contributing the most to the differences between stations through time (Table 11). Spionidae, capitellidae, phyllodocidae, and maldanidae are polychaete families, and nuculidae is a family of deposit-feeding bivalves.

The environmental variables best correlating to the difference in macrofaunal community composition through time were yearly average DIN and ammonium, $\rho = 0.165$. The second-best combination of variables was yearly average DIN, ammonium, and temperature, $\rho = 0.158$ (Table 11). The BEST analysis with a longer timeline and more abundance data resulted in a lower correlation (ρ), but mirrored the results of the BEST analysis from the previous comparison, which revealed nitrate + nitrite, temperature, and dissolved oxygen as the best correlated environmental variables (Table 10). Chlorophyll a was also tested, but was not among the most correlated environmental variables.

Table 11: Comparison 4 analysis results. Statistical analysis comparing current study macrofauna to all historical core studies. The test statistic R measures the difference between the average of rank similarities among replicates *within* sites and the average of rank similarities from pairs of replicates *between* sites. R=1 if *all* replicates within sites are more similar to each other than *any* replicates from different sites. The p (%) value is the percent significance level (Clarke 1993).

Test	Group	R	p (%)
ANOSIM	Between stations	0.534	0.1
	Between years	0.892	0.1
SIMPER	Species	PR average abundance	NJ average abundance
Between stations	Spionidae	18.76	11.05
	Capitellidae	14.48	14.08
	Phyllodocidae	6.77	4.49
	Nuculidae	4.19	9.69
	Maldanidae	3.23	7.71
BEST	Stressor variables	Rho	
	NH ₄	0.139	
BEST	Stressor variables	Rho	
	NH ₄	0.118	
	Temperature		

5: HISTORICAL FULL MACROFAUNAL COMPARISON

Providence River taxonomic distinctness appeared to vary with time, increasing through the 1970s, then decreasing in the early 2000s, then increasing again to the 2020s. This suggests a decrease in biodiversity of the Providence River benthic community assemblage in the early 2000s (Figure 14). North of Jamestown appears to decrease in taxonomic distinctness overall through time, peaking in the mid-1980s with a steady decrease through the 2020s. Rhode Island Sound, despite the lack of historical sampling, seems to display an overall decrease in taxonomic distinctness (Figure 14).

Abundance data from all historical studies, converted to presence/absence, and aggregated to the family level were originally plotted in MDS, but no clear patterns were revealed. To compensate for the difference in macrofaunal sampling methodology among historical studies, taxonomic distinctness was employed for the temporal comparison of all historical studies at all stations (Figure 14).

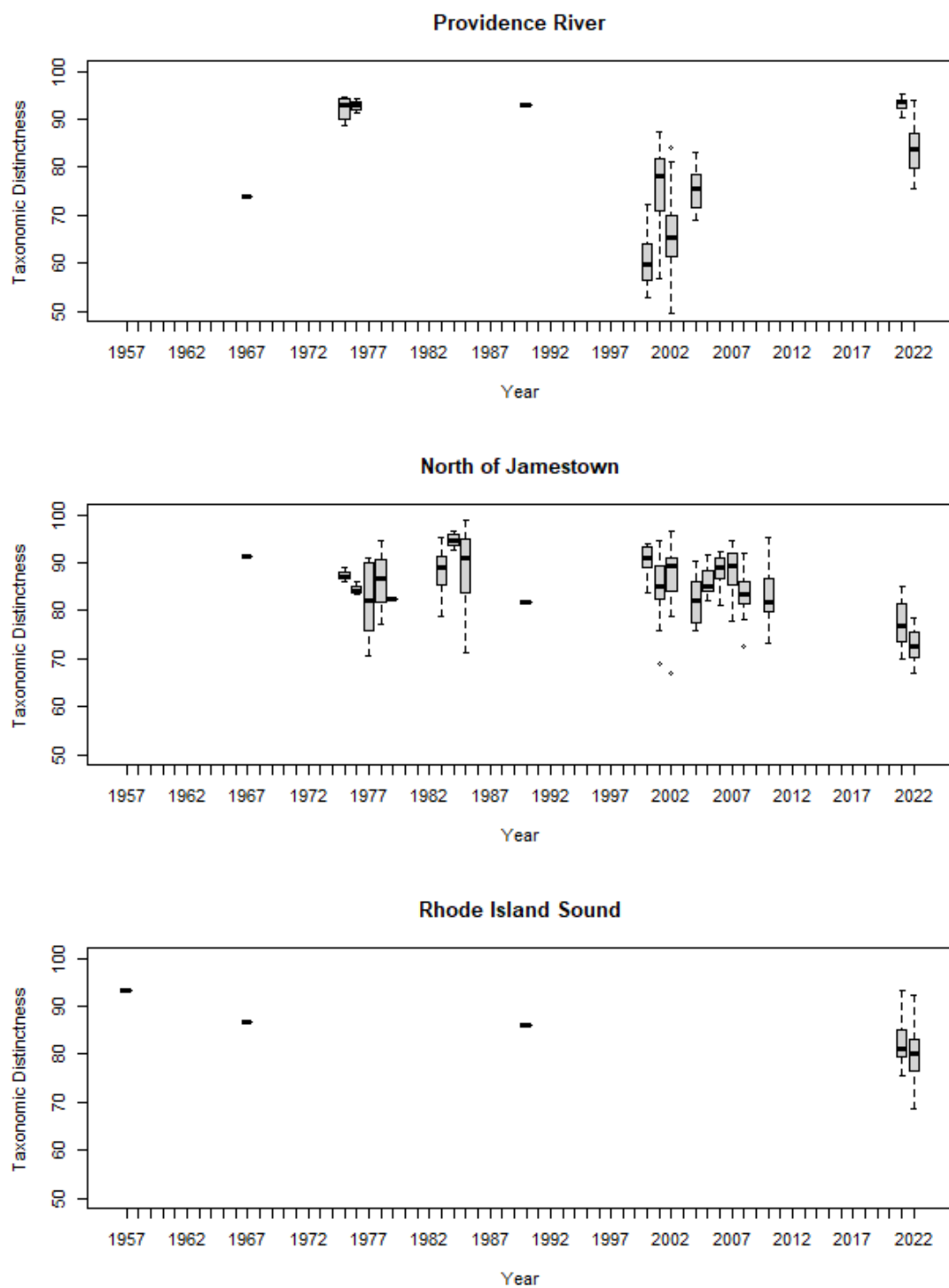


Figure 14: Time series of taxonomic distinctness (Δ^*) for each station, including all benthic historical studies, and replicates (Table 1).

DISCUSSION

We hypothesized that changes in environmental variables through time would change the macrofaunal community composition of the upper Bay through time. This hypothesis was supported by the results of the current study. BEST analysis revealed changes in nitrogen, dissolved oxygen, and temperature correlated the most with changes in macrofaunal abundance. The upper Bay Providence River macrofaunal community historically presented as a disturbed, low successional community. The current study found the upper Bay community to be less disturbed and more similar to the mid Bay North of Jamestown community through time. Between 2021 and 2022, the upper Bay Providence River station experienced a large decrease in both total abundance and total biomass. The three main objectives of the study were fulfilled with the five comparisons performed to test the hypothesis.

1: CURRENT MACROFAUNAL COMPARISON

The three stations sampled along the West Passage of Narragansett Bay presented as three distinct communities (Figure 7, Table 6). All presented high abundances and biomasses of the most common species found throughout the Bay, including *Mediomastus ambiseta*, *Nucula proxima* (previously *annulata*), *Nephtys incisa*, and *Mercenaria mercenaria* (Tables 4, 5). Historically, the North of Jamestown site was described as a *Nephtys incisa*-*Nucula proxima* community, an assemblage commonly found in Long Island Sound, Buzzards Bay, and other estuaries throughout the northeast

(Hale et al. 2018b). *Nephtys incisa*, commonly known as the catworm, is a large, long-lived, deep bioturbating, deposit feeding polychaete tolerant to disturbance. It dominated biomass of all three stations, with low abundance, described as a 'sub-dominating' species of the North of Jamestown site in the early 2000s (Table 5) (Calabretta & Oviatt 2008). The high biomass of *Nephtys incisa* and the ABC curves of all three stations presenting as less disturbed, as well as the temporal MDS result revealing the increasing similarity of the Providence River upper Bay site to the North of Jamestown mid Bay site suggested that the upper Bay, at least near Conimicut Point, increased to a higher successional stage (Table 5, Figures 6, 11, 12).

Macrofaunal successional stage, temporally or spatially moving away from an environmental disturbance, follows a normal trend transitioning from anoxic, to hypoxic, to normoxic sediment conditions and the community compositions normally found in such conditions (Pearson & Rosenberg 1976). Just beyond the source of pollution, sediment begins recolonization by small, highly abundant, tube-dwelling, tolerant polychaete species such as capitellids, spionids, and paraonids. Moving further from the disturbance, the sediment and its infaunal community becomes transitory, increasing the biodiversity of polychaetes and introducing small bivalves. Normoxic sediment hosts the richest biodiversity, increasing bivalve representation, introducing larger epifauna, and introducing larger, longer-lived, deeper bioturbating, and rarer species (Pearson & Rosenberg 1976). The increased similarity to the North of Jamestown station and the biomass dominance of the deeper bioturbating

Nephtys incisa, suggested that Providence River became less transitory and less disturbed in terms of benthic succession.

Despite communities at all three stations in both years presenting as less disturbed, there was a large discrepancy in not only the community composition of the Providence River station year-to-year, which may on its own be explained by year-to-year variability, but there was also a large decrease in both total replicate biomass and total replicate abundance of the upper Bay station (Figures 4, 6, 7). The upper Bay year-to-year community separation was also found in the previous study, comparing similar year-to-year changing, opportunistically-driven communities throughout three upper Bay stations to the North of Jamestown control station, which stayed stable year-to-year (Calabretta and Oviatt 2008). The previous study found prominent bottom water hypoxia at Conimicut Point. Dissolved oxygen concentration was found through BEST analysis to be one of the significant environmental variables driving station-to-station community differences, as well as year-to-year differences, but dissolved oxygen concentration at the Providence River station was not hypoxic at the time of sampling or directly before sampling (Tables 2, 6, 10).

Dissolved oxygen concentration at the Providence River station was hypoxic in late summer of 2021, when seasonal hypoxia is at its peak. Comparing the number of days when dissolved oxygen concentration was less than the hypoxic threshold of 3 mg/L, revealed that during the late summer of 2021, there were 50 days reaching less than 3 mg/L of dissolved oxygen, with

23 days reaching less than 2 mg/L, while in the late summer of 2020, there were only 37 days reaching less than 3 mg/L, with 11 days reaching less than 2 mg/L (NBFSMN 2021-2022). Late summer is also the time of macrofaunal larval development, settlement, and recruitment into the adult population, following the peak abundance period of June (Rudnick 1984). The sharp decrease in abundance and biomass at the Providence River station between 2021 and 2022 was probably due to decreased larval settlement and recruitment caused, in part, by extended exposure to stressful, hypoxic conditions. Late-summer defaunation events were observed in similar temperate estuaries throughout the world, including Chesapeake Bay, Tokyo Bay, Chinhae Bay, the Baltic Sea, and Scandinavian fjords (Lim 2006, Seitz et al. 2009, Kodama 2012, Gammal et al. 2016, Norkko et al. 2019). Delayed recolonization occurred in late fall and early winter of both Tokyo Bay and Chinhae Bay, but at lower abundance and biomass, similar to the 2022 samples of the current study.

In the early 2000s and before, bottom water hypoxia was mostly driven by nitrogen increasing organic matter input to the bottom and producing an overactive benthos that was limited in oxygen consumption by stratification. The hypoxic event in late summer of 2021 appears to mostly be driven by high rainfall. Total rainfall for the month of July 2021 was 7.12", while the normal monthly total is 2.91" (National Weather Service 2021). For comparison, July 2020 experienced 1.67" of rainfall, and July 2022 experienced 0.46". Both August and September 2021 also experienced higher rainfall than the normal

monthly average. Increased rainfall increases the amount of freshwater input to the Bay, increasing stratification while decreasing mixing of atmospheric oxygen to the bottom. Combining a wet July with a large biomass of epifaunal predators, and their respiration requirements, could decimate the population of a macrofaunal community between years, as well as greatly increase hypoxia in the area. Summer precipitation has been significantly correlated with summer days of hypoxia (Oviatt et al. 2017). In 2013, summer rainfall reached 42 cm, while the Bay experienced 242 days of summer hypoxia. In fact, 2007, considered the 'lower extreme' of summer hypoxia and rainfall, saw 20 cm of summer precipitation and 50 days of hypoxia, similar to the hypoxia experienced in summer of 2021, suggesting 2021 is at the lower extreme of summer hypoxia (Oviatt et al. 2017).

Epifaunal predator biomass was tested in the BEST analysis for the comparison between current stations and years of macrofaunal abundance, but was not found to be one of the variables correlating the most to the change in macrofaunal abundance (Table 6). The second set of variables correlating the most to change in macrofaunal abundance was sediment percent nitrogen and epifaunal biomass, however ($\rho = 0.825$). Both epifaunal biomass and macrofaunal abundance at the Providence River station decreased greatly between 2021 and 2022 (Figure 4, Figure 8). Epifaunal predators, including *Crangon septemspinosa*, feed on the macrofauna residing in the sediment. While presenting as most abundant at the Rhode Island Sound station in the beam trawl tows for epifauna, *C. septemspinosa* was also

caught in the diver-collected cores, presenting as the most abundant at the Providence River station.

Multiplying average abundance of *C. septemspinosa* caught in the macrofaunal cores in 2021 at the Providence River station (442 individuals/m²) and a daily estimated energy consumption of 130 J/day for a shrimp of 0.5 g at 20°C (average biomass of *C. septemspinosa* in the current study was 0.205 g, and instantaneous bottom temperature on the day of sampling was 19.8°C), yields 57,460 J/m²d or 57.46 KJ/m²d (Taylor & Peck 2004). The total macrofaunal energy found at the Providence River station in 2021, excluding *C. septemspinosa* and large bivalves, yields 1,363,129 J/m² or 1363 KJ/m². The estimated population of *C. septemspinosa* at the Providence River station in 2021 could consume that amount of energy in just 23 days, and that is considering only one of the epifaunal predators present in the assemblage. The oxygen consumption of an individual active, or feeding, *C. septemspinosa* at 14°C is estimated to be 6 mg O₂/day or 0.006 g O₂/day (Taylor & Peck 2004). Multiplied by the average PR abundance of the shrimp yields 2,652 mg O₂/day or 2.652 g O₂/day. Combining the large oxygen demand of a high biomass epifaunal community feeding on a high abundance macrofaunal community with stratification from increased freshwater input could explain the prolonged hypoxia and the large decrease of macrofaunal abundance, and epifaunal biomass, between years at the Providence River station.

2: CURRENT EPIFAUNAL COMPARISON

Epifaunal community composition mirrored the trend of macrofaunal community composition, presenting as three distinct communities following the environmental gradient of the West Passage (Figures 10, 11). The environmental variables best correlating to the differences in community composition between stations were temperature and salinity (Table 9). While this is a direct reflection of the natural temperature and salinity gradient within the estuary, what is more telling is the lack of significant correlation in epifaunal communities with carbon and/or nitrogen sediment content. Epifauna live on the sediment, but are not directly impacted by the content of the sediment like macrofauna. Large, mobile predators have been found to be most impacted by hypoxic conditions, but are also able to move away from stressful environments (Riedel et al. 2021). At the time of sampling, dissolved oxygen concentration was greater than 3 mg/L; late summer hypoxia had not set in yet. Given their mobility, most epifauna probably leave the area long before hypoxia becomes lethal, so dissolved oxygen would not be a significant environmental variable driving instantaneous epifaunal abundance at the time of sampling.

The large separation in replicate tows for each year are due to the lack of replication. Macrofaunal core sampling took 15 replicate cores per station each year, but only two replicate beam trawl tows were performed per station each year due to the large size of the transects towed and time constraints. Whereas the cores were taken in the general area of a predetermined

coordinate within the Bay for each station, the beam trawl transects were not as efficiently monitored or recorded. Start time, end time, average boat speed, and starting and stopping coordinates were recorded. Area towed was calculated from beam width, average boat speed, and time towed. Starting and stopping coordinates are not good measures of distance towed because the boat was not moving in a straight line either year and may have crossed over the same patch multiple times. Area towed around the selected core coordinates also varied greatly between stations and years as the spread of the tow area was not well controlled, meaning multiple benthic habitats may have been sampled that do not directly match that of the station cores.

Additionally, Narragansett Bay hosts ephemeral reefs of *Crepidula fornicata*, a gastropod commonly known as the slipper shell (NBNERR 2009). The shelf within the gastropod shell is perfect for spionid polychaete habitation, specifically *Polydora cornuta*. The beam trawl may sample a shifting reef one year, but not the next, even if sampling location is kept constant, representing an aggregate, partially mobile, episodic community, as was the case at the North of Jamestown station in 2022. A 1.5-cm mesh size with a 0.5-cm liner means that some mobile organisms less than 1.5 cm or 0.5 cm may have escaped from the net as it dragged along the bottom. Interestingly, the mesh net size did not impact the amount of *Crangon septemspinosa* sampled, which was also caught in macrofaunal core sampling, and a large portion of the macrofaunal biomass, when storing the overlying water at the sediment-water interface with the top 2 cm of sediment. *C. septemspinosa* was dragged up

tangled in mats of algae, especially at the Rhode Island Sound site. Mats of algae, bacteria, aggregations of sediment tubes, and reefs of shell hash were artifacts of beam trawl sampling that caused overlap with macrofaunal sampling due to small-sized organisms being brought up with the epifaunal growths.

Rhode Island Sound, the deepest station, was sampled with a 100-foot winch wire the first year, which was not long enough to drag the beam along the bottom for the entirety of the tow. This was corrected for the second year with a longer, 250-foot winch wire, resulting in a better 2022 sampling of Rhode Island Sound, but with a discrepancy in the much less abundant sampling of 2021. Another discrepancy is the date of sampling. In 2021, beam trawl sampling was performed on June 29th and 30th, the same days as core sampling. In 2022, due to diver availability, beam trawl sampling was performed on June 15th, while core sampling was done on June 29th. Still encompassing the peak June abundance, a two-week sampling gap sampling different parts of the same benthic community instantaneously will impact the data, especially if a hypoxic event, storm event, or predatory feeding event changed the abundance and biomass of the macrofaunal community sampled later.

3: BEFORE/AFTER TERTIARY NITROGEN MACROFAUNAL COMPARISON

Temporally, from the early 2000s to present, the Providence River community assemblage became more similar to the North of Jamestown community (Figure 12). A significant ANOSIM difference between Providence River stations before and after supported the community assemblage shift (Table 10). Environmental variables of nitrate and nitrite, temperature, and dissolved oxygen best correlated to the temporal changes of community composition. This supported the result of the current macrofaunal comparison, which found evidence of an increase in successional stage in the upper Bay, and is probably due to tertiary nitrogen removal, which specifically targeted the removal of DIN (Figure 6). The significant correlation with dissolved oxygen was probably also a result of tertiary nitrogen removal, as the decrease in organic matter falling to the benthos decreased bottom water respiration and increased bottom water dissolved oxygen. The time series for dissolved oxygen does reflect an increase in dissolved oxygen through time at the Providence River station, showing a stark recovery from severe hypoxia, even in the average summer signal (Figure 3c). Bottom temperature of the upper Bay increased more through time than the mid Bay bottom temperature, perhaps indicative of a shallower upper Bay (Figure 3a). Increased bottom temperature, seasonal freshwater stratification, and shallow depth increase the risk of hypoxia in the upper Bay, despite the decrease in nitrogen, the primary environmental driver, through time.

In the early 2000s, number of species at the Piverrovidence R, or Conimicut Point, station was less than 40 until 2004 when number of species, and total abundance, peaked. The total abundance at the Conimicut Point station in 2004 more than doubled compared to previously sampled years, coinciding with 93 days of hypoxia, beginning in May and ending in mid-October (NBFSMN 2004, Calabretta & Oviatt 2008). This high abundance parallels the total abundance experienced in 2021 at the PR station, albeit less than half the almost 250,000 individuals/m² experienced at Conimicut Point in 2004, which coincided with 50 days of hypoxia in late summer. Peak total abundances of macrofaunal communities and epifaunal communities may be driving prolonged hypoxic conditions due to increased respiratory demand from the increased number of individuals found per m².

Mediomastus ambiseta and *Nucula proxima*, the two most abundant species in the assemblages throughout the West Passage were also dominant throughout the early 2000s assemblages. *Ampelisca* was important in the assemblage of Mount Hope Bay, near Bristol, in the early 2000s, but not in the Conimicut Point assemblage, although it dominated the PR assemblage in 2021 (Calabretta & Oviatt 2008). *Polydora cornuta* (previously *ligni*) and *Streblospio benedicti* were dominant species found at Conimicut Point in the early 2000s and in the current study (Table 4). The oligochaete assemblage seems to be quite variable, only found in high abundance in 2004, and only found in high abundance in 2022, perhaps opportunistically taking advantage of deposit-feeding niches removed by hypoxia. The terebellid polychaete

Polycirrus medusa was a fairly numerous member of the North of Jamestown assemblage in the early 2000s, but was not found at all in 2021 or 2022 samples of the North of Jamestown station. It is a sessile tubeworm, considered second-order opportunistic, so it may have been out-competed, and predated, as the North of Jamestown station moved from transitory to a more mature, less disturbed successional stage and assemblage. Increased deposit-feeding paraonid and lumbrinerid polychaetes in the 2021 and 2022 North of Jamestown assemblages, such as *Paradoneis lyra*, *Levinsenia gracilis*, and *Scoletoma tenuis*, further indicated an increase in successional stage (Table 4) (Pearson & Rosenberg 1976). This succession suggests that although the impact of change in community composition is larger on the Providence River assemblage through time, the North of Jamestown assemblage may have also benefitted from tertiary nitrogen removal.

4: HISTORICAL CORE MACROFAUNAL COMPARISON

From the 1970s to present, macrofaunal community composition change correlated with DIN, ammonium, and temperature, mirroring the results of comparing before and after tertiary nitrogen removal, but with ammonium as a driver rather than nitrate and nitrite, probably due to the increased amount of data in the timeline assessed (Tables 10, 11). In 1984, a mesocosm experiment was performed to assess the recovery of the polluted ecosystem of Providence River when compared to control sites from the mid and lower Bay (Oviatt et al. 1984). The upper, mid, and lower Bay sites

sampled for the mesocosm experiment mirrored the sampling stations of the current study. Unfortunately, the data were not available for direct comparison. In the mesocosm experiment, the polluted Providence River ecosystem recovered after five months, becoming more similar to the assemblages of North of Jamestown and Rhode Island Sound, directly mirroring the results of the current study (Figures 12, 13). Although not on the same time scale, both the mesocosm experiment and the current study found the Providence River station became more similar to the North of Jamestown station, suggesting a decrease in disturbance and increase in successional stage. In the mesocosms, nutrient concentrations decreased, and oxygen concentrations increased, when compared to the non-treated, *in situ* polluted water of Providence River, again mirroring the results of the current study (Figures 2, 3).

The dominant species of the macrofaunal community are rather stable through time, with *Mediomastus ambiseta* and *Nucula proxima* as the top two most abundant species for most core studies examined in Table 1. In the Myers and Phelps (1978) core study performed in 1975-1976, *Turbonilla interrupta*, a pyramid shell gastropod very sensitive to disturbance, was the second most dominant species in the North of Jamestown assemblage after *Nucula proxima*. *T. interrupta* was not a dominant species in any of the three station assemblages, but between 2021 and 2022 at the Providence River station, *T. interrupta* greatly decreased in average abundance, probably due to the prolonged exposure to seasonal hypoxia in late summer of 2021, similar to the decrease in *Acteocina canaliculata* (Table 4). While there was a decrease

in abundance due to hypoxic exposure, an organism very sensitive to disturbance that historically was a large portion of the abundance at the North Jamestown assemblage further supports the recovery of Providence River from being considered a highly polluted site in the 1980s to an assemblage resembling more closely the mid Bay in the 2020s.

Acteocina canaliculata is a Carolinian gastropod with a climate trajectory moving northward (Hale 2017). Increased surface temperature in the Bay has caused a shift in species composition and increased abundance of more southern species, including a decrease in eel grass abundance, a shift from winter flounder and other boreal demersal fish to more southern, smaller pelagic fish, an increase in decapods, an increase in *Crassostrea virginica*, or the Eastern oyster, and an increase in several polychaete, mollusk, and gastropod species, most of which are found in Narragansett Bay (Oviatt 2004, Hale 2017).

A SIMPER analysis was also performed to compare assemblage changes through time by splitting the timeline being analyzed into three phases, the 1970s and 1980s as phase 1, the early 2000s as phase 2, and the 2020s as phase 3. Comparing phase 1 to phase 3 shows the largest change in time. *Mediomastus ambiseta*, the most abundant species found in the Bay increased in average abundance between time phases. *M. ambiseta* is another species presenting a northern shifting climate trajectory, as is *Tharyx acutus*, a cirratulid polychaete that increased in average abundance between phase 1 and phase 3 (Hale 2017). *Mulinia lateralis*, the dwarf surf clam, has a

southern shifting climate trajectory, and found a decrease in average abundance between time phases.

5: HISTORICAL FULL MACROFAUNAL COMPARISON

Taxonomic distinctness, indicative of biodiversity, varied greatly through time at the Providence River station, probably reflecting the opportunistic communities, and the station's proximity to WWTF (Figure 14). It may also be attributed to dissolved oxygen concentration. Organisms sensitive to hypoxia may have been absent from the assemblage in the early 2000s when dissolved oxygen was low, but recovered in the 2020s when bottom dissolved oxygen concentration returned to normoxic levels. The variation in taxonomic distinctness at North of Jamestown may have also been explained by increase in dissolved oxygen, but North of Jamestown never reached the low oxygen experienced in the upper Bay (Figure 3). Given the significant correlation with nitrate + nitrite in the BEST analysis of before and after tertiary nitrogen removal (Table 10), the increase in biodiversity from 2000-2010, followed by leveling off in the 2020s, was probably indirectly due to tertiary nitrogen removal. The decrease in biodiversity at the Rhode Island Sound station, especially the impact of the low value in 2022, may have been due to the lack of historical sampling to compare against, but decrease in biodiversity in the lower Bay may also be attributed to the removal of tertiary nitrogen. Down the Bay, nutrient concentrations end up decreasing oligotrophic production rather than eutrophic over-production (Nixon et al. 2008). Past study data revealed a slight, although not significant, increase in taxonomic distinctness through time,

encompassing all of Narragansett Bay (Hale 2018b). The slight increase may reflect the variation in taxonomic distinctness at varying sites throughout the Bay, as seen in the current study (Figure 14). Bay-wide benthic biodiversity seems to be increasing in response to the decrease in tertiary nitrogen removal and, more directly the increase in bottom dissolved oxygen, but individual variabilities at different sites throughout the Bay responding to the complex of environmental variables constantly impacting the benthic community may be decreasing the significance.

COMPARISON TO OTHER ESTUARIES

The long-term recovery of Narragansett Bay from eutrophication and hypoxia was similar to the recovery of other estuaries recovering from similar stressors. From 1992 to 2006, Boston Harbor benthic habitats shifted in succession from anaerobic to aerobic assemblages in response to improved WWTF, similar to the response in the upper Bay station of the current study (Diaz et al. 2008). Boston Harbor experienced an initial increase in *Ampelisca* abundance and tube mat concentration in 1992, declining by 2005, indicative of shifts in successional stage assemblage from a more opportunistically-driven assemblage that included *Ampelisca* to a more transitory to normal assemblage including more deposit feeders and tube-dwelling polychaetes outcompeting the opportunistic *Ampelisca*. Narragansett Bay experienced high concentrations of *Ampelisca* tube mats in Mount Hope Bay near Bristol in the early 2000s. The current Providence River station experienced a high

abundance of *Ampelisca* in 2021, subsequently removed from the assemblage by epifaunal predators and seasonal hypoxia exposure in late summer 2021, and was much less abundant in 2022 (Calabretta & Oviatt 2008). This change suggests that Providence River is probably currently shifting away from and opportunistic successional assemblage, and heading toward a decline in *Ampelisca* abundance in the future.

On a smaller size scale, the Gullmarsfjord, a small fjord with a maximum depth of 118 m in Sweden, recovered from a severe anoxia event that eliminated all fauna (Rosenberg et al. 2002). Over two years, the successional stage of the fjord increased from opportunistically-driven to a mature, normal benthic assemblage. The exact path of successional stage decrease was not retraced by the observed successional increase, rather, taking an independent pathway to a mature successional stage in the context of the environmental variables experienced at the time of recovery. This mirrors not only the successional recovery of Narragansett Bay, but most benthic successional recoveries as successional recovery is not linear, but contextual (Duarte et al. 2009). The rapid recovery is due to the small size of the fjord, which also showed that the deeper the depth, and the larger the scale, the slower the recovery, echoing the rapid, 5-month, recovery of the small mesocosms from the polluted Providence River (Oviatt et al. 1984, Rosenberg et al. 2002). The recovery of a large estuary like Narragansett Bay took much longer and is still currently recovering.

From 2001-2004, severe hypoxia decreased benthic diversity, biomass and abundance of Chesapeake Bay (Seitz et al. 2009). Two hypoxic events have been identified in Chesapeake Bay, an early summer hypoxia trend and a late summer hypoxia trend (Murphy et al. 2011). The late summer hypoxia was attributed to nitrogen loading, while the early summer hypoxia was attributed to stratification. Due to an apparent decrease in nitrogen loading to Chesapeake Bay reported by the US Geological Survey, late summer hypoxia decreased. Due to climate change and large-scale climactic forcing, early summer stratification strength has increased during June from 1985-2009, increasing early summer hypoxia (Murphy et al. 2011). In Narragansett Bay, reduced nutrient loading decreased the length of seasonal hypoxia; in the early 2000s it began in April or May, but increasing rainfall due to climate change is increasing the length and severity of seasonal hypoxia again.

Modeling techniques implemented in the Baltic Sea were used to show future projections of benthic response to climate change and different nutrient loading scenarios (Ehrnsten et al. 2020). Due to climate change, even with reduction in nutrient loading, the model predicted a decrease in overall macrofaunal biomass and carbon processing by the end of the century. In the high nutrient load scenario, biomass and carbon processing increased to the mid-21st century, but then rapidly decreased. The study overall showed that the benthic response to environmental change is nonlinear, and increasing climate change will weaken coastal benthic-pelagic coupling (Ehrnsten et al. 2020). The results of the current study suggest the benthic community of

Narragansett Bay is responding to other environmental variables, such as epifaunal predator biomass and hypoxia induced by stratification caused by increased freshwater input, despite the decrease in nutrient reduction.

LOOKING FORWARD

Benthic response to environmental variables is complex and contextual. While it may seem moot to decrease nutrient loading if summer stratification caused by increased rainfall is just going to replace it as the cause of bottom water hypoxia, nutrient loading can be directly mitigated by management, whereas climate change, and its many impacts, cannot. Rather than a two-fold anthropogenic impact on benthic communities, decreasing organic matter feeding the increased metabolism caused by increased temperatures may decrease the pace of stratification-induced hypoxia. There are many nuances and indirect relationships involved in sediment nutrient fluxes and benthic macrofauna that are understudied. Oxygen was found to be the main driver of benthic communities and nutrient fluxes, including ammonium and phosphate, with macrofauna and meiofauna as significant predictors (Norkko et al. 2019). In the North Sea, bioturbation and bioirrigation, with temperature, were the most important variables influencing oxygen and nutrient flux (Neumann et al. 2021). Eutrophic nutrient loading and stratification can cause hypoxia and decrease the succession of benthic communities, but macrofauna are interacting with sediment in ways not fully understood that may shed further

light on management or man-made mitigation techniques that improve bottom water dissolved oxygen.

CONCLUSION

Change in nitrogen as an environmental variable correlated the most through time with changes in the upper Bay macrofaunal community. Changes in dissolved oxygen and temperature also correlated with the change in the upper Bay macrofaunal community. However, epifaunal biomass and rain-fall driven stratification appear to be causing late summer seasonal hypoxia currently, impacting the biomass and abundance of the upper Narragansett Bay macrofaunal community 8 years after the completion of tertiary nitrogen removal. The three communities sampled throughout the West Passage of Narragansett Bay present as three distinct, healthy macrofaunal assemblages. The historically polluted upper Bay Providence River assemblage has become more similar through time to the mid-Bay North of Jamestown assemblage, with the appearance of species high in biomass and sensitive to disturbance, suggesting a high successional stage. Despite uncontrollable climate change, and resulting environmental variables, the results of the current study represent successful improvement of macrofaunal successional stage, abundance, and biomass, in a historically polluted community through managed nutrient reduction.

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